抓斗桥式起重机
Grab Bridge Crane

使用、操作、维修手册
Use, Operation and Maintenance Manual

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目录
Table of Contents

第一章 概述
Chapter I Overview

本使用维护说明书适用于固体垃圾搬运用控制抓斗起重机（以下简称起重机）。
This manual applies to control grab cranes (hereinafter referred to as cranes) for solid waste handling.

用户应严格遵照本“说明书”的规定操控，使用和维护起重机。
The user shall operate, use and maintain the crane strictly in accordance with the provisions of this "Manual".

标准件，外购件的使用、维护请按供货商提供的资料进行（请查阅“相关资料”）。
For the use and maintenance of standardized parts and outsourced parts, please follow the information provided by the supplier (please refer to "Related Information").

1.1 用途
1.1 Purpose

1.1.1 垃圾抓斗起重机，位于垃圾贮存坑的上方，主要承担垃圾的投料、搬运、取物等工作。
1.1.1 The waste grab crane, located above the waste storage pit, is mainly responsible for waste feeding, handling and removal.

1.1.2 投料：当进料口的垃圾不足时，起重机抓取垃圾运行至进料口上方，进料漏斗加料。
1.1.2 Feeding: When there is insufficient garbage in the feeding port, the crane grabs the waste and runs above the feeding port, and the feeding funnel starts feeding.

1.1.3 搬运：将靠近卸料门的垃圾运到贮坑的其他地方，避免卸料的拥堵，调节坑内垃圾数量。
1.1.3 Handling: Transport waste close to the discharge gate to other parts of the waste storage pit to avoid congestion of the discharge and to regulate the amount of waste in the pit.

1.1.4 取物：将不慎进入贮坑中，但不宜处理的物体取出。
1.1.4 Removal: Remove objects that have inadvertently entered the pit but are not suitable for handling.

1.2 使用条件
1.2 Use Condition

1.2.1 起重机的电源为三相交流，频率 50HZ，电压 380V。电机和电器上的允许电压波动的上限为额定电压的+10%，下限（尖峰电流时）为额定电压的-15%。
1.2.1 The power supply of the crane is three-phase AC with frequency 50HZ and voltage 380V. The upper limit of permissible voltage fluctuations on motors and appliances is +10% of the rated voltage, and the lower limit (in case of peak currents) is -15% of the rated voltage.

1.2.2 起重机安装地点的海拔高度不超过 1000m。
1.2.2 The altitude of the crane installation site cannot be over 1000m.

1.2.3 环境温度-20~+500℃，在 24 小时内的平均温度不超过+350℃。电气控制设备，尤其是电子设备应保证在恒温环境工作（温度范围：00~+400℃）。
1.2.3 Environment temperature is -20 - +500℃, and the average temperature cannot be over +350℃ during a 24-hour period. Electrical control equipment, especially electronic equipment, should be guaranteed to work in environment with a constant temperature (temperature range: 00 - +400℃).

1.3 主要技术参数
1.3 Main Technical Parameters

1.3.1 结构形式：电动双梁抓斗桥式起重机。
1.3.1 Structure form: electric double-beam grab bridge crane

1.3.2 起重机的具体参数：生产率、起重量、跨度、工作级别等见起重机总图。
1.3.2 The specific parameters of the crane: productivity, lifting capacity, span, working level, etc. are shown in the crane general drawing.

1.3.3 控制方式：采用主令控制器、按钮、触摸屏配合控制屏（柜、箱）的控制系统。
1.3.3 Control mode: adopt a system that combined master controller, buttons, and touch screen with control screen (cabinet, box).

1.3.4 操纵方式：手动（操纵人员通过联动台操纵起重机完成移动、抓斗升降、抓取、投料等动作）和半自动（操纵人员选取和投放位置并输入重复的次数，起重机自动从泊车位置启动、移向抓取点、下降抓斗、抓取垃圾，提升抓斗、移向进料口，称重计量、投料、返回泊车位置重复动作）控制方式。
1.3.4 Control mode: manual control mode (the operator manipulates the crane by the linkage console to complete the movement, grab lifting and lowering, grabbing, feeding and other actions) and semi-automatic control mode (the operator selecting the position to feed and entering the number of times to repeat, the crane automatically starts from the parking position, moves to the grabbing point, lowers the grab, grab the waste, and then lifts the grab, moves it to the inlet, weighs and meters, feeds, and returns to the parking position to repeat).

1.4 起重机的安装与调试
1.4 Installation and Debugging of the Cranes

1.4.1 安装准备及注意事项：
1.4.1 Installation preparations and precautions:

起重机安装人员在进场开始安装之前，应会同委托安装单位及制造单位的代表一起开箱，按照随机所带的装箱单，清点核对交货物与装箱单所列的零部件数量是否相符，有无短缺和损伤，随箱文件是否齐全，然后应检查所有文件和金属结构外观有无损坏，对照安装架设附加图和有关技术文件中的技术要求，认真研究安装方案和安装架设的具体程序。

Crane installation personnel should open the box with the representatives from entrusted installation unit and manufacturer before installing in the field. With the packing list randomly brought, the installation personnel needs to count and check whether the quantity of the goods and the parts listed in the packing list is consistent with each other, whether there are any shortages and damages, whether the accompanying documents are complete or not, and then they should check whether there are any damages to all the documents and the appearance of the metal structure, and carefully study the installation program and the specific procedures for installation and erection in comparison with the additional diagrams of the installation and erection and the technical requirements in the relevant technical documents.

起重机卸车搬运，应特别注意避免发生起重机受到扭、弯、撞击等事故，起吊时至少须有两个吊点，吊点的捆扎处须有衬垫物，捆扎以走轮或主梁本身为限，并尽可能选在桥架的两端即主梁与端梁的连接处，存放时应安置平稳，并用枕木放平垫实，安装前应对搬运不当和存放不好造成的缺陷和超过规定误差的部分，按技术要求调整修复，对金属结构部份的缺陷，必须在地面得到校正，否则不准架设。
Special attention should be paid to avoid accidents such as crane twisting, bending, strike and so on when unloading and handling. There must be at least two lifting points when lifting, and the bundling place of the lifting points must have padding. Bundling is limited to the walking wheels or the main girders themselves and is selected, wherever possible, at the ends of the bridge frame, i.e., where the main beams are connected to the end beams. It should be placed in a smooth when storing, and put flat and solid with sleepers. Before installation, defects caused by improper handling and storage, as well as parts exceeding specified error should be adjusted and repaired according to technical requirements. Defects in the metal part of the structure must be corrected at ground level, or the erection will not be permitted.

1.4.2 桥架的吊装桥架是通用桥式起重机的主体，其架设和组装都有严格的技术要求，并有一套严格的测量方法来测量安装结果。
1.4.2 Lifting bridge is the main body of the general bridge crane, and there are strict technical requirements for its erection and assembly, as well as a set of strict measurement methods to measure the installation results.

1.4.3 安装方法与要求
1.4.3 Installation methods and requirements

A）垫桥架的要求：
A) The requirements for padding the bridge:

对于起重量≤75 吨的起重机，一般采用地面组装整体架设。地面组装时应将起重机支承在按起重机跨度临时搭起的架子上，使主梁离开地面。如果架子上可以铺设钢轨，则大车车轮可直接放在钢轨上，否则应把支架垫在主梁的下盖板靠近主梁两端变高处。无论采用哪种方法垫，均需用水平仪找正。铺钢轨的以钢轨上平面为基准找水平；垫在主梁下的，在端梁宽度中心线上，距小车轨道中心线如表 2-1 的 A、B、C、和 D4 点上测量桥架的水平。在测量水平位置之前，应将端梁按图样规定的方法（用铰制孔螺栓或用铰轴）连接好。有锥形踏面的车轮，应使钢轨的侧面与轮缘接触，以免影响水平的准确性。
For cranes with a lifting capacity of ≤75 tons, ground assembly is generally used for overall erection. Ground assembly shall be done by supporting the crane on a

temporary frame based on the span of the crane to make sure that the main beam is off the ground. If rails can be laid on the frame, the wheels of the crane can be placed directly on the rails, otherwise the bracket should be padded on the lower cover of the main beam near where the ends of the main beam are high. It needs to be aligned with a gradienter no matter what kind of padding method is used. Padding with rails finds the alignment based on the plane of the rails. For padding under the main beams, measure the level of the bridge at the centerline of the width of the end beams, at points A, B, C, and D4 from the centerline of the trolley tracks as shown in Table 2-1. Before measuring the horizontal position, the end beams should be attached as specified in the drawings (with reamed hole bolts or with reamed shafts). Wheels with tapered treads should have the side of the rail in contact with the rim in case of the inaccuracy of the level.

B）组装桥架的质量要求：
B) Quality requirements for assembling the bridge:

支承点设在车轮下的桥架，可按表 2-1 的规定，逐项检查桥架的安装质量，如是垫在主梁下，则除主梁上拱度一项外，其余项目均可检查，同时还可以检查大车运行机构的安装质量。
Supporting point located under the bridge of the wheels, can be checked the installation quality of the bridge item by item in accordance with the provisions of Table 2-1. For example, for padding under the main beam, in addition to the camber of the main beam, the rest of the items can be inspected, and the quality of the installation of the crane running mechanism can be checked too.

1.4.5 小车的安装：
1.4.5 The installation of trolley

中小吨位的标准双梁小车已经组装完成并经过试运行，经检测和必要的调整，即可装在桥架上。
Standard double-beam trolleys of small and medium tonnage have been assembled and test run and are ready to be mounted on the bridge after testing and necessary adjustments.

1.4.6 大车运行机构：
1.4.6 Crane running mechanism

大车运行机构已由制造厂组装在桥架上，并试运转合格。大车运行机构在起重机组装完后，应根据下列要求进行安装检查：
The crane running mechanism has been assembled on the bridge frame by the manufacturer and tested to be qualified. The crane running mechanism shall be installed and inspected according to the following requirements after assembling:

A）检查基准：检查运行机构时，应以主动轮外侧为基准。
A) Inspection standard: When inspecting the running mechanism, the outside of the driving wheel should be used as the reference.

B）车轮端面的水平偏斜如图 2-2，每个车轮的水平偏斜 P≤1/1000L（L 为测量长度），且两个主动（或被动）车轮的不平行方向应相反，如图 2-3 所示的任何一种组合形式都是符合要求的。
(B) Horizontal deflection of the wheel end as shown in Figure 2-2, the horizontal deflection of each wheel $P \leq 1/1000L$ (L stands for the measured length), and the

direction of the non-parallelism of the two driving (or driven) wheels should be opposite, as shown in Figure 2-3, any one of the combinations of the form is in line with the requirements.

C）测量跨度：见表 2.1
C) Measurement span: see Table 2.1

D）车轮找正：车轮找正包括平行度、垂直度及同位差。
D) Wheel alignment: Wheel alignment includes parallelism, perpendicularity and tolerance.

a.平行度：用 0.6～1.0mm 钢丝，在车轮下部拉上两条平行线，看车轮内外侧面与平行线接触的情况，如果平行线在车轮 4 个侧面上都均匀接触，这说明前后车轮平行并在同一条中心线上；如果有间隙，则测量一下尺寸是否在规定的范围内，若不在规定的范围内，应松开固定角轴承座的螺栓，进行调整，必要时，可把固定角型轴承箱的垫板铲下来或用气割割下来，待车轮调好后再焊。
a. Parallelism: Using 0.6-1.0mm steel wire to pull two parallel lines on the lower part of the wheel and to observe the contact between the inner and outer sides of the wheel and the parallel line. If the parallel lines are in uniform contact with all four sides of the wheel, it indicates that the front and rear wheels are parallel and on the same centerline; If there is a gap, measure whether the size is within the specified range. If not, loosen the bolts that fix the angle bearing seat and adjust it. If necessary, shovel off the base plate of the fixed angle bearing box or use gas cutting to cut it off. Wait for the wheels to be adjusted before welding.

b.垂直度：车轮在垂直方向偏斜不应大于 1/400L（L 为测量长度），且上部应向外（如图 2—4）。
b. Perpendicularity: The deviation of the wheel in the vertical direction should not exceed 1/400L (L is the measured length), and the upper part should be outward (as shown in Figure 2-4).

c．同一端梁下车轮的同位差（图 2 一 5）：车轮同位允差：2 个车轮时≤2mm,3 个或 3 个以上时≤3mm。
c. The tolerance of the wheels under the same end beam (Figure 2-5): The tolerance of the wheels:≤ 2mm for 2 wheels, ≤ 3mm for 3 or more wheels.

测量方法：在端梁外侧拉一钢丝，平行于端梁中心线，其高度接近于车轮的轮缘，测量点在车轮的垂直中心线上，测量出钢丝与各轮测量点之距离，选定一个车轮为基准，则其他车轮的同位差即可计算出来。
Measurement method: Pull a steel wire on the outer side of the end beam, paralleling to the centerline of the end beam, with a height close to the wheel flange. The measurement point being on the vertical centerline of the wheel, the distance between the steel wire and the measurement points of each wheel should be measured. Select a wheel as the reference, and the tolerance of other wheels can be calculated.

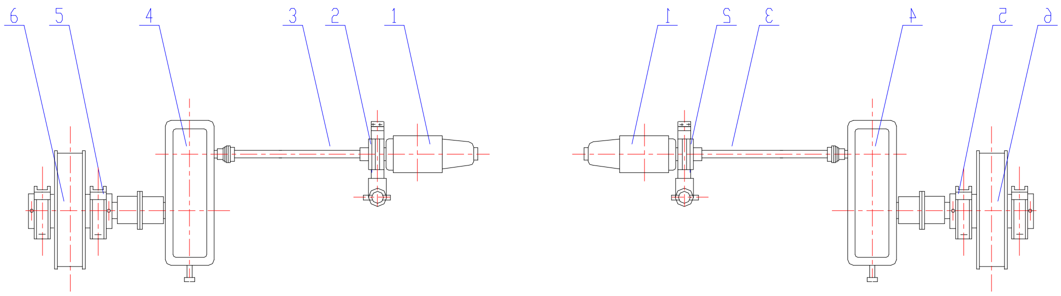


图 1-1 大车运行机构
Fig. 1-1 Crane Running Mechanism

1、电动机；2、制动器；3、传动轴；4、减速机；5、轴承箱；6、车轮
1. Motor; 2. Brake; 3. Transmission shaft; 4. Reducer; 5. Bearing box; 6. Wheels

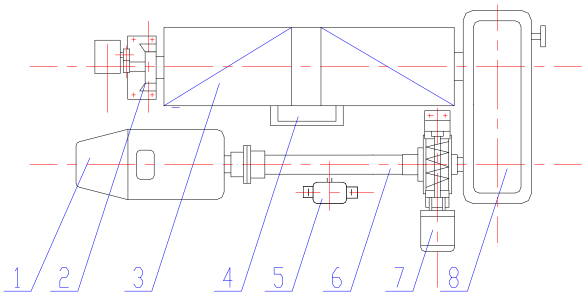


图 1-2 起升机构
Fig. 1-2 Lifting Mechanism

1、电动机；2、旋转起升安全开关；3、卷筒组；4、定滑轮；5、重锤起升安全开关；6、传动轴；7、制动器；8、减速机
1. Motor; 2. Rotating lifting safety switch; 3. Drum set; 4. Fixed pulley; 5. Heavy hammer lifting safety switch; 6. Transmission shaft; 7. Brake; 8. Reducer

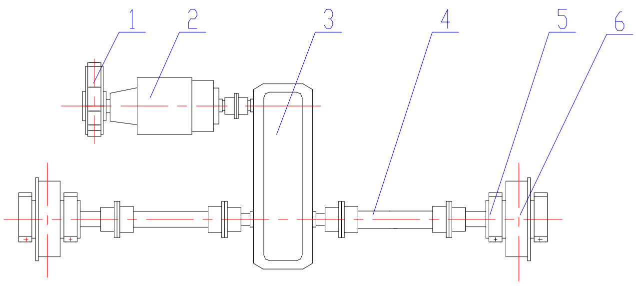
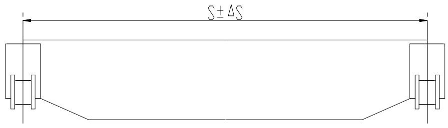
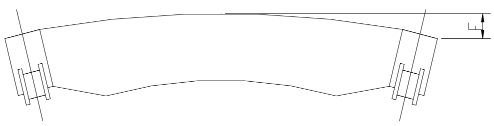
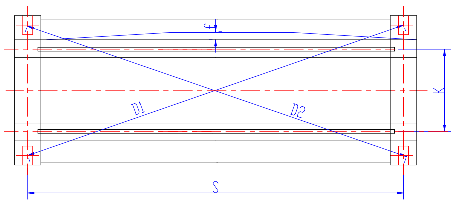
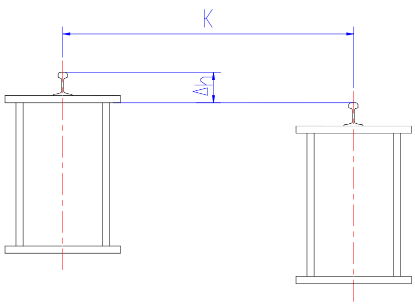
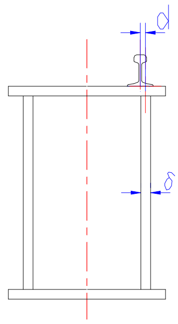


图 1-3 小车运行机构
Fig. 1-3 Trolley Running Mechanism

1、制动器；2、电动机；3、减速机；4、传动轴；5、轴承箱；6、车轮
1. Brake; 2. Motor; 3. Reducer; 4. Transmission shaft; 5. Bearing box; 6. Wheels

表 2-1
Table 2-1

序号 S/N	项目 Item	简图 Sketch	允许偏差（mm） Allowable Deviation (mm)
1	由车轮量出的跨度偏差 Span deviation measured by wheels		△S=5 且主动轮的跨度值与被动轮的跨度值之差不得大于 5mm △ S=5, and the difference of the span value between the driving wheel and the driven wheel should not exceed 5mm
2	装配后主梁的上拱度 Camber of the main beam after assembly		F=（0.9/1000～1.4/1000）S F=(0.9/1000 - 1.4/1000)S
3	桥梁对角线偏差｜D_1-D_2| Diagonal deviation of bridge |D1-D2|		｜D_1-D_2｜≤5 对称箱形梁：f≤1/2000S,且当起重量≤50t 时只许向走台侧凸曲 Symmetrical box girder: f ≤ 1/2000S, and only the lifting weight is ≤ 50t, protruding towards the walkway is allowed
4	主梁水平旁弯度（走台和端梁都装上后） Horizontal side camber of the main beam (after installing the platform and end beam)		
5	小车轨距偏差 Trolley gauge deviation		对称箱形梁： Symmetrical box girder: 跨端处：士 2 End span: ±2 跨中处：S≤19.5m，+5 +1 Mid span: S≤19.5m，+5 +1 S＞19.5m，+7 +1 其他梁：士 3 Other beams: ±3 K≤2m，△h≤3 2m＜K＜6.6, △h≤0.0015K K≥6.6，△h≤10
6	同一截面小车轨道高低差△h Height difference △h of trolley tracks of the same section		

7	小车轨道中心线与轨道梁腹板中心线的位置偏差 Position deviation between the centerline of the trolley track and that of the track beam web		偏轨箱形梁： Bias-rail box girder δ＜12,d≤6 δ≥12,d≤δ/2

1.4.7 起重机的运转试验，是新安装架设的起重机所必须进行的检验项目，也可作为定期检验的项目。
1.4.7 The operation test of the crane is a necessary for newly installed and erected cranes, and can also be a regular inspection.

试车前的准备和检查
Preparation and inspection before test run

为保障运转试验安全、顺利地进行，试验前必须对起重设备进行认真检查，并为试验做好充分准备。
To ensure the safe and smooth operation of the test, the lifting equipment must be carefully inspected before the test and fully prepared.

A）关闭起重机电源，检查所有连接部位的紧固情况。
A) Turn off the power supply of the crane and check the tightness of all connecting parts.

B）各传动机构装配是否精确灵活，金属结构有否变形，钢丝绳在滑轮和卷筒上的缠绕情况是否牢固。
B) Check whether the assembly of each transmission mechanism is accurate and flexible, whether the metal structure is deformed, and whether the winding of the steel wire rope on the pulley and drum is firm.

C）用兆欧表检查电路系统和所有电气设备的绝缘电阻是否符合要求。
C) Use a megohmmeter to check whether the insulation resistance of the circuit system and all electrical equipment meets the requirements.

D）在断开动力线路的情况下，检验操纵线路接线的正确性，检查所有操纵设备的传动部分是否灵活可靠。
D) When disconnecting the power circuit, verify the correctness of the wiring of the control circuit and check whether the transmission parts of all control equipment are flexible and reliable.

E）各润滑点加注润滑油脂，减速机、制动器的液压缸、液压缓冲器等，按规定加润滑油。
E) Add lubricating grease to each lubrication point, and add lubricating oil to the reducer, hydraulic cylinder of brake and hydraulic buffer according to regulations.

F）保证电气设备工作正常可靠，其中必须特别注意电磁铁、限位开关、安全开关和紧急开关的工作可靠性，注意分别驱动进行机构电动机的接线相序检查，使两电动机同向运转。

F) To ensure the normal and reliable operation of electrical equipment, especially the reliability of electromagnets, limit switches, safety switches, and emergency switches, it is important to drive separately and check the wiring phase sequence of the mechanism motor to ensure that the two motors operate in the same direction.

G）加强人员的安全防护，清除大车运行轨道上、起重机上以及试验区域内影响运转试验的一切物品。
G) Strengthen the safety protection of personnel and remove all items on the running track of the crane, on the crane and in the test area that affect the running test.

H）准备好负荷试验的重物，重物可用相对密度比较大的钢锭、钢坯、条材、生铁块或大型铸造毛坯等。但必须要对质量计算准确，并应可靠捆扎好。
(H) Prepare well weights for load test, which can be those with a relatively high density, such as steel ingots, billets, bar stocks, pig iron blocks or large casting billets, etc. However, the quality must be accurately evaluated and should be reliably bundled.

空运转试验
Idling test

A）小车行走：空载小车沿轨道往返运行 3 次，此时，不应有明显打滑。主动车轮应在轨道全长上接触，起动制动应正常可靠，限位开关的动作准确，小车上的缓冲器与桥架上的撞头相碰的位置准确。
A) Trolley travel: The empty trolley travels back and forth along the track three times, and there should be no obvious slipping at this time. The driving wheels should be in contact with the entire length of the track, the starting and braking system should be normal and reliable, the action of the limit switch should be correct, and the position of the buffer on the trolley and the collision head on the bridge should be accurate.

B）抓斗空载升降：开动小车起升机构，使空钩上升、下降三次，此时起升机构限位开关的动作应准确可靠。
B) Grab no-load lifting: Start the trolley lifting mechanism to raise and lower the unloaded hook three times. At this time, the limit switch of the lifting mechanism should operate accurately and reliably.

D）把小车开到跨中，起重机沿厂房全长行走两次，以验证房架和轨道，然后以额定速度往返行走三次，检验行车机构的工作质量。此时启动或制动时，车轮不应打滑，行走平稳；限位开关的动作准确；缓冲器工作正常。
D) Driving the trolley to the mid span, the crane will travel twice along the entire length of the factory building to verify the frame and track. Then, it will travel back and forth at the rated speed three times to verify the working quality of the crane mechanism. At this point, when starting or braking, the wheels should not slip and the travel should be smooth; The action of the limit switch is accurate; and the buffer is working properly.

E）抓斗供电电缆的收、放速度应与钢丝绳卷筒速度相协调。
E) The speed of reeling up and out of the grab service cable should be coordinated with the speed of the wire rope drum.

载荷起升能力试验
Load lifting capacity test

空运转试验试车情况正常之后，才允许进行载荷起升能力试验，载荷起升能力试验分静载试验和动载试验两种，先进行静载试车，再进行动载试车。
The load lifting capacity test can only be conducted after the idle running test is normal. The load lifting capacity test can be divided into dead load test conducted at first and dynamic load test.

A）静载试验
A) Dead load test

静载试验的目的是检验起重机及其各部分结构的承载能力。
The purpose of the dead load test is to verify the load-bearing capacity of the crane and its various parts of the structure.

首先起升较小的负荷（可为额定起重量的 0.5～0.75 倍）运行几次，然后起升额定负荷，在桥架全长往返运行数次后，将小车停在桥架中间，定出测量基准点，起升 1.25 倍额定负荷，离开地面约 100～20Omm 左右，停悬 10min，卸去负荷，分别检查起升负荷前后量柱上的刻度（在桥架中部或厂房的房架上悬挂测量下挠度用的线锤，相应的在地面或主梁上安设一根量柱），反复试验最多 3 次桥架不得产生永久变形（即前后两次所检查的刻度值相同）,然后将小车开至跨端，检查实际上拱度值应在 0.7S/1000～1.4S/1000 范围内；最后使小车停在桥架中间，起升额定负荷，离地面 100mm 左右，测量主梁的挠度，其挠度不应超过 S/800。
First, lift the smaller load (which can be 0.5-0.75 times of the rated lifting weight) and run it several times, then lift the rated load. After running back and forth several times throughout the entire length of the bridge, stop the trolley in the middle of the bridge, set the measurement reference point, and lift 1.25 times of the rated load. Leave the ground about 100-200mm, suspend for 10min, remove the load, and check the scale on the measuring column (plumb bob hung in the middle of the bridge or on the building frame of the factory or for measuring lower deflection, with a measuring column installed on the ground or main beam accordingly) before and after lifting the load, repeat the test for up to 3 times, and the bridge should not produce permanent deformation (i.e. the scale values checked before and after are the same), and then drive the car to the end span to check that the actual camber value should be within the range of 0.7S/1000-1.4S/1000; Finally, stop the trolley in the middle of the bridge, lift the rated load, and measure the deflection of the main beam about 100mm above the ground. The deflection should not exceed S/800.

B）动载试验
B) Dynamic load test

动载试验的目的主要是验证起重机各机构和制动器的功能。
The main purpose of dynamic load test is to verify the functions of various mechanisms and brakes of the crane.

起升 1.1 倍额定载荷作为动载试车，分别开动各机构（也可同时开动 2 个机构），试验中，对每种动作应在其整个运动范围内反复启动和制动，并按其工作循环与工作级别，试车时间应延续 1 小时，试验后，起重机各功能正常，没有发现机构或结构损坏，连接处无松动或损坏现象，即为合格。
Lift the rated load for 1.1 times as a dynamic load test, and start each mechanism separately (or simultaneously start two mechanisms). During the test, each action should be repeatedly started and braked within its entire range of motion, and the test

should continue for 1 hour according to its working cycle and class. After the test, as long as the crane's functions are normal, and there is no damage to the mechanism or structure, and no looseness or damage at the connection, then the test is considered qualified.

1.5 构造简介

1.5 Brief Introduction of the Structure

1.5.1 桥架：

1.5.1 Bridge:

采用偏轨箱形主梁与端梁采用“不等高”的联接而成，端梁和主梁分别运输。小车轨道采用轨道方钢与主梁焊接。

Adopting bias-rail box main beam and end beam, through "unequal height" connection, the two being transported separately. The trolley track is welded with the main beam using track square steel.

1.5.2 大车行走机构：

1.5.2 Crane walking mechanism:

采用分别驱动布置，将车轮组和“三合一”驱动单元安装在端梁上，“三合一”由硬齿面减速器和制动电机组成。从动车轮轴上装有位置编码器，当大车运行时编码器轴随之运行，编码器生成的信号传入 PLC，由 PLC 调用功能块进行数值运算，最终将大车具体的运行距离显示在触摸屏上。

Adopting a separate drive arrangement, the wheel set and the "3-in-1" driving unit are installed on the end beam. The later consists of a hardened face reducer and a brake motor. A position coder is installed on the driven wheel axle. When the crane runs, the coder shaft runs accordingly. The signal generated by the coder is transmitted to the PLC, and the PLC works on numerical calculation through the function block. Finally, the specific running distance of the crane is displayed on the touch screen.

1.5.3 起升机构：

1.5.3 Lifting mechanism:

一台电机通过两台硬齿面减速器驱动两个卷筒上的四根钢丝绳吊起四吊点液压抓斗，四绳间距上宽下窄，使抓斗具备防摆特性，停车定位准确。减速器与卷筒采用卷筒联轴器联接，降低了对卷筒安装精度和小车架刚度的要求，并提高减速器低速轴密封的寿命，装拆方便。电缆卷筒采用无动力传输，确保钢丝绳与电缆线完全同步运行。

The motor drives four steel wire ropes on two drums through two hardened face reducers to lift a hydraulic grab with four lifting points, and the distance between the four ropes is wide at the top and narrow at the bottom, making the grab have anti-swing characteristics and accurate parking positioning. The reducer and the drum are connected by the drum coupling, which reduces the requirements of the drum installation accuracy and the stiffness of the trolley frame, and increases the sealing life of the low-speed shaft of the reducer, and is convenient to install and disassemble. The cable drum adopts unpowered transmission to ensure that the steel wire rope and the cable run completely in synchronously.

1.5.4 抓斗机构：
1.5.4 Grab mechanism:

抓斗采用四吊点多瓣液压抓斗，核心液压系统采用原装进口产品（电磁阀切换），采用 380V 交流电压驱动。液压缸位于抓斗外侧，便于检修，且每个液压缸上都安装防护罩。抓斗供电采用一体式单层缠绕电缆卷筒。（注：抓斗电缆缠绕卷筒时设防护措施、在电缆松动时不允许电缆在缠绕卷筒时向卷筒两侧串动掉在传动部位上造成故障）。
The grab adopts four-point multi-lobe hydraulic grab, the core hydraulic system adopts original imported products (solenoid valve switching), and uses 380V AC voltage drive. The hydraulic cylinder is located on the outer side of the grab for easy maintenance, and each hydraulic cylinder is equipped with a protective cover. Grab power supply adopts integrated single-layer winding cable drums. (Note: Protective measures should be taken when a grab cable is wound around the drum. When a cable is loose, it is not allowed for the cable to move towards both sides of the drum and fall on the transmission part, causing faults.)

1.5.5 小车运行机构：
1.5.5 Trolley running mechanism

采用“三合一”传动装置，采用分别驱动布置，将车轮组和“三合一”驱动单元安装在端梁上，“三合一”由硬齿面减速器和制动电机组成。从动车轮轴上装有位置编码器，当大车运行时编码器轴随之运行，编码器生成的信号传入 PLC，由 PLC 调用功能块进行数值运算，最终将小车具体的运行距离显示在触摸屏上。
Adopting the "three-in-one" transmission device, and installing the wheel set and the "three-in-one" drive unit on the end beam. The "three-in-one" is composed of a hardened face reducer and a brake motor. The position encoder is installed on the driven wheel axle. When the crane starts, the encoder shaft operates accordingly. The signal generated by the encoder is transmitted to the PLC, which calls the function block for numerical calculation. Finally, the specific running distance of the trolley is displayed on the touch screen.

1.5.6 起升高度限制器：

1.5.6 Lifting height limiter:

DXZ 型，安装在卷筒轴上。限位器与卷筒同轴运行，分为下限位，上减速限位，上限位，上极限限位。当到达相应的调节好的限位位置时起升机构可以减速或停止。

DXZ type, installed on the drum shaft. The limit switch operates coaxially with the drum, and is divided into lower limit switch, upper deceleration limit switch, upper limit switch, and upper extreme limit switch. When reaching the corresponding adjusted limit position, the lifting mechanism can slow down or stop.

1.5.7 称量装置：

1.5.7 Weighing devices:

起重机上安装有称重传感器，并通过有线传递信号到控制室，计量打印报表。称重系统同时具备超载报警功能，载荷达到额定起重量的 90%时，发出提示性报警；载荷达到额定起重量的 110%时，发出报警并停机。称重显示、计量打印系统使用工控计算机进行设置和管理，可按用户要求输出各种管理报表。

The crane is equipped with a weighing sensor, and transmits signals to the control room through wires to measure and print the report. The weighing system also has overload alarm function, when the load reaches 90% of the rated weight, the warning alarm will be issued. When the load reaches 110% of the rated lifting weight, an alarm will be issued and the machine will shut down. The weighing display and metering printing system uses the IPC to set and manage, and can output various management reports according to user requirements.

1.5.8 运行机构行程限位开关：
1.5.8 Stroke limit switch of running mechanism:

采用感应式传感器作为限位装置，以感应铁元件为基础，感应距离大约2.5cm~3cm。可以有效避免由老式的传感器感应距离短或者机械式限位接触不良引起的故障，可以降低故障几率和维修。
Based on sensing iron elements, an inductosyn is used as a limiting device, and the sensing distance is approximately 2.5cm-3cm. It can effectively avoid faults caused by outdated sensors with short sensing distances or poor mechanical limit contacts, and can reduce the probability of faults and maintenance.

1.5.9 供电：
1.5.9 Power supply:

大、小车采用电缆滑车供电系统。
The crane and the trolley use cable pulley supply system.

大车供电工字钢在车间牛腿侧面，小车供电工字钢在主梁侧面。

The crane power supply I-steel is on the side of the bracket of the workshop, and the trolley power supply I-steel is on the side of the main beam.

1.5.10 控制方式：
1.5.10 Control modes:

电控操作集中在固定的操纵室内，采用手动控制和半自动控制方式。
The electronic control operation is centralized in the fixed control room, using manual control and semi-automatic control modes.

1.6 电气设备
1.6 Electrical Equipment

本起重机的控制采用具有强大逻辑运算功能的可编程序控制器（简称 PLC），传

动使用交流变频调速，可使起升、小车、大车（简称：三机构）运行平稳，准确停车。

The control of this crane adopts a programmable logic controller (PLC) with powerful logic operation functions, and the transmission uses AC variable-frequency speed regulation, which can ensure smooth operation and accurate parking of the lifting mechanism, trolley, and crane (referred to as the three mechanisms).

1.6.1 动力电源：

1.6.1 Power supply:

380V，三相五线，50HZ，所需容量：见电气总图。经柜内总空开 F1，总接触器 KM1 送出，分配到起升机构三相保护开关 F11、小车运行机构三相保护开关 F14、大车运行机构三相保护开关 F16 及抓斗开闭机构电机三相电机保护器开关 F10 等各机构。总空气开关 F1 具有欠压脱扣功能，脱扣机构的激磁线圈电压过低时机构分励。

380V, three-phase-five-wire, 50HZ, and the required capacity: see General Electrical Diagram. Through the total air switch F1 in the cabinet, the total contactor KM1 is sent out and distributed to the three phase protection switch F11 of the lifting mechanism, the three phase protection switch F14 of the trolley running mechanism, the three phase protection switch F16 of the crane running mechanism and the three phase motor protection switch F10 of the grab motor opening and closing mechanism. The total air switch F1 has the undervoltage trip function, and when the excitation coil voltage of the trip mechanism is too low, the mechanism will shunt.

1.6.2 控制电源：

1.6.2 Control power supply:

220V，单相三线，50HZ。取自电源柜的保护开关 F2，经系统中的控制变压器 T1 分配到各个机构。

220V, single-phase-three-wire, 50HZ. The protective switch F2, taken from the power cabinet, is distributed to each mechanism through the control transformer T1 in the system.

1.6.3 信号及计算机电源：

1.6.3 Signal and computer power supply:

信号（各机构限位开关，接触器控制等）电源由保护开关 F4 提供。PLC 的电源由保护开关 F5 提供。

The signal (limit switch of each mechanism, contactor control, etc.) power supply is provided by the protection switch F4. The power supply of the PLC is provided by the protection switch F5.

1.6.4 照明电源：

1.6.4 Lighting power supply:

交流 220V，单相，50HZ，取自平移柜内的保护开关 F9。

AC 220V, single-phase, 50HZ, taken from the protection switch F9 in the translation cabinet.

1.6.5 辅助电源：

1.6.5 Auxiliary power supply:

柜内照明灯、柜内风扇由保护开关 F7 提供。
The lighting and fan inside the cabinet are provided by the protection switch F7.

1.6.6 电铃电源：
1.6.6 Bell power supply:

警示电铃电源由保护开关 F9 提供。
The power supply for the warning bell is provided by the protection switch F9.

1.6.7 中央控制设备：
1.6.7 Central control equipment:

起重机的中央控制设备为“中央控制柜”，是起重机的控制中心，安装在电气室。柜内配置可编程序控制器（简称 PLC）一组；有电源模块、数字量输入模块、数字量输出模块、高速计数器模块等组成。起重机上各种行程、限位开关的信号，各机构主令控制器的指令信号及各机构的辅助信号，信号进入柜内由输入继电器进行电压隔离和转换，变为直流 24V 信号送入 PLC，经过程序处理对各机构进行有序的控制。
The central control equipment of the crane is the "central control cabinet", which is the control center of the crane and is installed in the electrical room. A set of programmable logic controller (PLC) configured in the cabinet is composed of power module, digital input module, digital output module, high-speed counter module, etc. The signals of various stroke and limit switches on the crane, the command signals of the main controller of each mechanism and the auxiliary signals of each mechanism, the signals enter the cabinet by the input relay for voltage isolation and conversion, into the DC 24V signal into the PLC, and the orderly control of each mechanism is processed by the program.

1.6.8 起升工作原理：
1.6.8 Working principles of lifting:

起升采用全程磁通矢量电流控制、速度开环变频调速线路，使起升机构在上升、下降时可获得稳定的速度。动力电源通过开关和继电设备进入变频器，在变频器内由整流单元整流，将交流电变成直流电。经过逆变单元将直流电变为交流电，逆变单元的输出频率全程可控，电动机可以获得不同频率的交流电源、起动、运行在不同的速度上。当电动机的转速超过同步转速（变频器的频率值）时，电动机进入回馈状态，其电能返回变频器，变频器的制动单元自动将这些能量泄放在制动电阻器上，使电动机获得制动转矩。装置具有过压、过流、超速、缺相等保护。
The lifting mechanism adopts a full range flux vector current control and a speed open-loop variable-frequency speed control circuit, allowing it to achieve stable speed during lifting and lowering. The power supply enters the frequency converter through switches and relay equipment, and is rectified by the rectifier unit inside the frequency converter, converting AC power into DC power. Through the inverter unit, the DC is changed into AC, the output frequency of the inverter unit is controllable, and the motor can obtain AC power of different frequencies, start, and run at different speeds. When the speed of the motor exceeds the synchronous speed (frequency value of the converter), the motor enters a feedback state, and its electrical energy returns to the converter, and the brake unit of the converter automatically drains this energy onto the brake resistor, so that the motor obtains the braking torque. The device has protections

of over voltage, over current, over speed, phase loss.

1.6.9 起升控制方式：
1.6.9 Lifting control modes:

起升调速装置可由操作员通过主令控制器进行控制。由触摸屏进行控制时，系统为自动控制状态。手动状态时，采用主令控制器控制，控制器的档位为 4-0-4，两个方向对称，1 档低速，变频器输出 10~15HZ，2 档中速，变频器输出 25HZ，3 档高速，变频器输出 35HZ，4 档全速变频器输出 50HZ。上述两种控制方式的信号输入 PLC，根据需要由程序发出运行指令，用来调整变频器的频率，对起升电机进行控制。
The lifting speed regulating device can be controlled by the operator through the master controller. When controlled by the touch screen, the system is in automatic control state. In the manual state, the master controller is used to control, the gear of the controller is 4-0-4, the two directions are symmetrical, the first gear is low speed, the converter output 10-15HZ, the second gear is medium speed, the converter output 25HZ, the third gear is high speed, the converter output 35HZ, and the fourth gear is full speed converter output 50HZ. The signals of the above two control modes are input into the PLC, and the running instructions are issued by the program according to the needs, which are used to adjust the frequency of the converter and control the lifting motor.

1.6.10 抓斗工作原理：
1.6.10 Working principles of the grab:

抓斗采用电动液压系统，液压泵由一台电动机驱动，当 F10 上电后，在 PLC 指令的控制下闭合电动机正向转动抓斗打开，闭合电动机反向转动抓斗闭合。
The grab adopts the electro-hydraulic system, the hydraulic pump is driven by a motor, when F10 is powered on, under the control of PLC instructions to close the motor to open the grab with forward rotation, and to close the grab with reverse rotation.

1.6.11 抓斗控制方式：
1.6.11 Grab control modes:

抓斗开闭可由操作员终通过主令控制器进行控制。采用主令控制器控制，控制器的档位为 1-0-1，两个方向对称。
The opening and closing of the grab can be controlled by operators through the master controller. The master controller is used to control, and the gear of the controller is 1-0-1, and the two directions are symmetrical.

1.6.12 平移运行机构控制设备：
1.6.12 Control equipment for translation running mechanism

运行机构采用全程磁通矢量电流控制、V/f 方式变频调速线路，使运行机构可获得稳定的速度。动力电源通过开关和继电设备进入变频器，在变频器内由整流单元整流，将交流电变成直流电。经过逆变单元将直流电变为交流电，逆变单元的输出频率是全程可控的，电动机可以获得不同频率的交流电源，电动机起动运行在不同的速度上。当电动机的转速超过同步转速（变频器的频率值）时，电动机进入回馈状态，其电能返回变频器，变频器的制动单元自动将这些能量泄放在制动电阻器上，使电动机获得制动转矩。大车、小车运行机构控制线路基本一致，主要区别：大车运行机构带动四台电动机，小车运行机构带动两台电动机，

The running mechanism adopts full flux vector current control and V/f frequency conversion speed regulating circuit, so that the running mechanism can obtain stable speed. The power supply enters the frequency converter through switches and relay equipment, and is rectified by the rectifier unit inside the frequency converter, converting AC power into DC power. The DC power is converted into AC power through an inverter unit, and the output frequency of the inverter unit is fully controllable. The motor can obtain AC power at different frequencies, and the motor starts and runs at different speeds. When the speed of the motor exceeds the synchronous speed (frequency value of the converter), the motor enters a feedback state, and its electrical energy returns to the converter, and the brake unit of the converter automatically drains this energy onto the brake resistor, so that the motor obtains the braking torque. The control lines of the crane and the trolley running mechanism are basically the same, the main differences are: the crane running mechanism drives four motors but the trolley running mechanism drives two motors,

1.6.13 平移运行机构控制方式：
1.6.13 Control modes of translation running mechanism:

由主令控制器进行控制。采用主令控制器控制，控制器的档位为 4-0-4，两个方向对称，1 档低速，变频器输出 10~15HZ，2 档中速，变频器输出 25HZ，3 档高速，变频器输出 35HZ，4 档全速变频器输出 50HZ。
It is controlled by the master controller. Using a master controller for control, the controller has a gear range of 4-0-4, symmetrical in both directions. The first gear is low speed, and the converter outputs 10-15HZ. The second gear is medium speed, and the converter outputs 25HZ. The third gear is high speed, and the converter outputs 35HZ. The fourth gear is full speed, and the converter outputs 50HZ.

1.6.14 联动台：
1.6.14 Linkage console:

联动台是起重机的操作中心，上述各机构的手动控制和自动控制均在联动台上。在联动台上可完成起重机电源的起动、停止、紧急停止、各机构运行信号的输等工作。
The linkage console is the operation center of the crane, and the manual control and automatic control of the above-mentioned mechanisms are on the linkage console. On the linkage console, the crane power supply can be started, stopped, stopped in emergency, and the transmission of operating signals of various mechanisms.

1.6.15 称量系统：
1.6.15 Weighing System:

称量系统由计算机、仪表和一台打印机组成。系统用来称量和记录投料口的投料数量，并与 PLC 进行信息交换参与完成自动控制。
The weighing system consists of a computer, a meter and a printer. The system is used to weigh and record the quantity of feeding port, and exchange information with PLC to complete automatic control.

1.6.16 电气线路图读图说明：
1.6.16 Reading instructions of the electrical circuit diagram:

电气线路图是逻辑草图，所有符号应符合电气制图标准和制造商－浙江浙起机械有限公司的相关电气图。

The electrical circuit diagram is a logical sketch, and all symbols should conform to the electrical drawing standards and the relevant electrical drawings of the manufacturer - Zhejiang Zheqi Machinery Co., Ltd.

第二章 使用与操作
Chapter II Usage and Operation

2.1 安全规则
2.1 Safety Regulations

2.1.1 应建立完善的安全生产责任制。准确了解并掌握设备的性能、特点，正确操作以保证人身设备的安全。应建立安全、技术档案。
2.1.1 A sound system of responsibility for production safety should be established. It is necessary to accurately and fully understand the performance and characteristics of the equipment, and operate the equipment correctly to ensure the safety of both the personnel and the equipment. Safety and technical archives should be established.

2.1.2 起重机操纵人员必须具有良好的视力和听力、身体健康，受过起重机的操作及起重机电气、机械设备方面维修、保养的学习和训练，并经安全技术部门考试合格者持证上岗。
2.1.2 Crane operators must have good vision and hearing, good physical health, have received learning and training in crane operation, maintenance and repair of crane electrical and mechanical equipment, and pass the safety and technical department examination to hold a certificate.

2.1.3 设备工作时应具备相关部门的准用证件，禁止设备带病工作。
2.1.3 When the equipment is working, it is necessary to have the relevant department's approval documents, and it is prohibited to work with equipment that is faulty.

2.1.4 起重机运行（包括自动控制）时，操作人员不得离开岗位，发现异常情况，应及时采取相应措施。
2.1.4 When the crane is working (including automatic control), the operator should not leave the post. If any abnormal situation is found, corresponding measures should be taken in time.

2.1.5 起重机工作时禁止进行清扫、加油等检修工作，禁止人员停留在起重机上。相关人员必须停留时，应采取有效的安全措施。
2.1.5 When the crane is working, cleaning, refueling and other maintenance work are prohibited, and personnel are prohibited to stay on the crane. When relevant personnel must stay, effective safety measures should be taken.

2.1.6 起重机工作完毕，必须切断总电源。在因检视或检修而停止工作时，应切断电源并悬挂醒目的警示牌。
2.1.6 After work, the main power supply of the crane must be cut off. When stopping work due to inspection or maintenance, the power should be cut off and conspicuous warning signs should be hung.

2.1.7 检修工具及备件必须贮存在专门的柜子内，禁止散放在起重机上。
2.1.7 Maintenance tools and spare parts must be stored in specialized cabinets, do not scatter on the crane.

2.1.8 检修时必须采用电压在 36V 以下的携带式照明灯。
2.1.8 The voltage of the portable lighting used in maintenance must be below 36V.

2.1.9 禁止用抓斗运送或起升人员。

2.1.9 It is prohibited to transport or lift personnel with a grab.

2.1.10 禁止将易燃（如煤油）、易爆、有毒物品贮存在起重机上。
2.1.10 It is prohibited to store flammable (such as kerosene), explosive, or toxic materials on cranes.

2.1.11 起重机大小车运行时避免抓斗刮碰垃圾、墙壁、料斗等以防止吊具旋转、碰撞。
2.1.11 When the crane and trolley are working, avoid scraping the grab against waste, walls, hoppers, etc., to prevent the hanger from rotating and colliding.

2.1.12 起重机操作人员工作期间不可饮酒，保证作业时精神集中，头脑清醒。
2.1.12 During work, crane operators are not allowed to drink alcohol, ensuring that they are focused and clearheaded.

2.2 起重机操作人员的职责
2.2 Responsibilities of Crane Operators

2.2.1.操作人员应熟悉该起重机的用途、操作规程、操作方法及保养知识。
2.2.1. Operators should be familiar with the use, regulations, methods and maintenance of the crane.

2.2.2 严格遵守安全规则、操作规程和维护保养规则。
2.2.2 Rules of safety, operation and maintenance should be strictly observed.

2.2.3 在主开关接电以后，所有控制器必须处于"零位"，检查起重机的情况。
2.2.3 After the main switch is powered on, all controllers must be in the "zero position" to check the condition of the crane.

2.2.4 起重机在每次开动以前，应发出警告信号（电铃）。操纵时无论手动还是自动操作，操作人员必须集中精力，注意关注抓斗的运行方向、速度和位置及抓斗的扭、摆情况，注意倾听设备尤其是电气设备的声音是否正常。特别注意：制动器的可靠性、操纵的稳定性、限位开关的可靠性等。
2.2.4 Before each start of the crane, a warning signal (electric bell) should be issued. Whether manual or automatic operation, the operator must concentrate on paying attention to the direction, speed and position of the grab and the twist and swing of the grab, and pay attention to listening to whether the sound of the equipment, especially the electrical equipment, is normal. Special attention: brake reliability, operating stability, limit switch reliability, etc.

2.2.5 吊起重物必须在垂直的位置。不允许利用大车及小车来拖拉重物。
2.2.5 Lifting heavy objects must be in a vertical position. Cranes and trolleys are not allowed to drag heavy objects.

2.2.6 起重机应以最缓慢的速度，逐步靠近边缘位置。正常工作时，禁止使用极限开关作为正常停车的手段。
2.2.6 The crane should gradually approach the edge at the slowest speed. During normal operation, it is prohibited to use limit switches as a means of normal stopping.

2.2.7 起重机的各控制器应逐级开动，在机械完全停止运转前，禁止将控制器从顺转位置直接反到逆转位置来进行制动，但在用作防止事故发生的情况下例外。

2.2.7 Each controller of the crane should be started step by step, and it is forbidden to brake the controller directly from the forward turning position to the reverse position before the machine completely stops operating, except in the case of preventing accidents.

2.2.8 操作人员要防止起重机与一台起重机相互碰撞。不可用一台起重机来移动另一台起重机。
2.2.8 Operators should prevent the crane from colliding with one another. One crane cannot be used to move another crane.

2.2.9 起重机的电动机突然停电或电压急剧下降时，应尽快将所有控制器拉回零位；总开关必须切断。
2.2.9 When the motor of the crane suddenly fails or the voltage drops sharply, all controllers should be pulled back to zero position as soon as possible; The main switch must be cut.

2.2.10 司机每班应对起重机进行检查。做好交接班（填与"交接班情况记录单"）。
2.2.10 The driver should inspect the crane every shift. Record the shift (fill in the "Shift Record Sheet").

垃圾吊车十不要
Ten Don'ts of the Waste Crane

1.大车到边、不要动小车；2.抓斗碰地、不要动吊车；3.两台行车、不要相互碰；4.操作作业、不要人走开；5.称重系统、不要私自调；6.抓斗到底、不要多放绳；7.发现问题、不要再使用；8.无证人员、不要乱开车；9.下班以后、不要忘断电。10.发现违章、不要不上报。
1. Move the crane to the side and don't move the trolley; 2. Don't move the crane when the grab touches the ground; 3. Don't collide with another crane when the two move at the same time; 4. Don't walk away while operating; 5. Don't adjust the weighing system without authorization; 6. Don't put more ropes when the grab is down to the bottom; 7. Don't use again If problems are found; 8. Don't drive without the license; 9. After work, don't forget to cut off the power. 10. Don't fail to report any violation you found.

2.3 操作步骤
2.3 Operation Steps

2.3.1"电源柜"
2.3.1 "Power Cabinet"

1）闭合空开—F1，
1) Closed air switch – F1,

2）闭合空开—F2，
2) Closed air switch – F2,

3）闭合空开—F3，
3) Closed air switch – F3,

柜内照明灯电源：本柜需要照明时，在柜门打开时灯亮。

Cabinet lighting power supply: When the cabinet needs to be lit, the light will be lit when the cabinet door is opened.

桥下照明灯总电源：需要桥下照明灯时，转动“桥下照明灯”开关（联动台右箱），桥下照明灯亮。
Overall power supply of underbridge lighting: When the underbridge lighting is needed, turn the "UNDERBRIDGE LIGHTING" switch (right box of linkage console), and the underbridge lighting will be on.

起重机上电源插座总电源：闭合检修内保护开关 F8。
Main power supply of the power socket on the crane: Close the maintenance internal protection switch F8.

注：电气系统的 PLC 应长期供电，没有必要不允许断电，必须停电时应首先断开 PLC 电源，然后依次断开其他机构的供电开关。
Note: The PLC of the electrical system should be powered on for a long time, and it is not allowed to power off if there is no need. When the power must be cut off, the PLC power should be disconnected first, and then the power switches of other mechanisms should be disconnected in sequence.

2.3.2“平移柜”
2.3.2 "Translation Cabinet"

大车运行机构合闸顺序：
Closing sequence of crane running mechanism:

1）闭合空开—F7，
1) Closed Air Switch – F7,

2）闭合空开—F8，
2) Closed Air Switch – F8,

3）闭合空开—F9，
3) Closed Air Switch – F9,

柜内需要照明时，在柜门打开时灯亮。
When the cabinet needs to be illuminated, the light comes on when the door is opened.

2.3.3“起升柜”
2.3.3 Lifting Cabinet

1）闭合动力电源空开 F11
1) Closed Power Supply Air Switch F11

2）闭合起升风机保护断路器 F12（在平移柜内）
2) Closed Lifting Fan Protection Circuit Breaker F12 (Within the Translation Cabinet)

3）闭合起升抱闸保护断路器 F13（在平移柜内）
3) Closed Lifting Brake Protection Circuit Breaker F13 (Within the Translation Cabinet)

柜内需要照明时，在柜门打开时灯亮。
When the cabinet needs to be illuminated, the light comes on when the door is

opened.

2.3.4“平移柜”
2.3.4 Translation Cabinet

小车运行机构合闸顺序：
Trolley Running Mechanism Closing Sequence：

1）闭合动力电源空开 F16，
1) Closed Power Supply Air Switch F16,

2）闭合制动器电源空开 F17，
2) Closed Brake Power Supply Air Switch F17,

大车运行机构合闸顺序：
Closing sequence of crane running mechanism:

1）闭合动力电源空开 F14，
1) Closed Power Supply Air Switch F14,

2）闭合制动器电源空开 F15，
2) Closed Brake Power Supply Air Switch F15,

抓斗机构合闸顺序：
Grab Mechanism Closing Sequence:

1）闭合电动机保护断路器 F10,
1) Closed Motor Protection Circuit Breaker,

柜内需要照明时闭合，在柜门打开时灯亮。
The cabinet is closed when light is needed inside, and the light comes on when the door is opened.

2.4 开车
2.4 Start-up

起重机按上述顺序完成合闸工作后进入“起动”过程：
After the crane completes the closing work in the above sequence, it enters the "Startup" process.

2.4.1 打开 PLC 柜内工作计算机，并使其进入正常的工作状态。
2.4.1 Switch on the working computer in the PLC cabinet and ensure it is in normal operating condition.

2.4.2 观察各机构的主令控制器手柄是否处于“零位”（手柄必须在零位）。
2.4.2 Observe whether the master controller handle of each mechanism is in the "zero position" (the handle must be in the zero position).

2.4.3 按动总电源起动按钮。
2.4.3 Press the start button of the main power supply.

2.4.4 三机构的起动顺序可互换。PLC、变频器起动时需要程序自检，其风机起动比较缓慢。通电后变频器未充分起动，立即起动机构对变频器不利，所以应在变

频器起动一段时间（大约 10 秒）再进行操作。起动顺序不对可能造成系统出现故障显示或报警。
2.4.4 The starting sequence of the three mechanisms is interchangeable. PLC and converter start-up needs to be programmed to self-test, and its fan start-up is slow. The converter is not fully started after powering on, and starting the mechanism immediately is unfavorable to the converter, so it should be operated after the converter has been started for a period of time (about 10 seconds). Incorrect starting sequence may cause the system to display a fault indication or alarm.

2.4.5 操作左箱手柄完成大车的左右行驶和小车前后行驶；操作右箱手柄完成抓斗的起升和开闭。起升、大、小车三个机构主令控制器的档位均为 4—0—4，方向对称。第一档为低速档，为停车准确而设立，正常运行不应长时间使用低速档。出于频繁点动尤其是正、反方向点动会使设备的使用寿命降低，所以操纵者应熟练掌握、充分利用本起重机的性能，准确、平稳地操作控制手柄，减少或不使用点动操作，控制吊重的摆动。禁止手柄快速、频繁的地点动或正反向切换或档位变换。
2.4.5 Operate the left box handle to complete the left and right travelling of the crane, and the forward and backward travelling of the trolley; and operate the right box handle to complete the lifting, opening and closing of the grab. The mechanism master controllers of the lifting, crane and trolley are all in 4-0-4 gears with symmetrical directions. The first gear is low-speed, set for accurate parking, and should not be used for a long time in normal operation. Because frequent jogging especially positive and negative jogging will reduce the service life of the equipment, so the operator should master and make full use of the crane's performance to operate the control handle accurately and smoothly, and reduce or not conduct jogging operation to control the lifting load swing. Rapid and frequent handle jogging or forward and reverse switching or gear shift are prohibited.

2.5 停车
2.5 Stop

2.5.1 停车：
2.5.1 Stop:

起重机长时间停止使用或需检修，应将起重机的电源全部切断，各种断路器恢复到断开位置（包括用户的起重机总断路器）。断电顺序：控制手柄放到“零位”；各机构停止运行后，按动联动台停止按钮断电，断开 F1 空开。必要时应断开平移柜和 PLC 柜的所有空开及称重装置的电源。
When the crane is out of service for a long time or needs to be overhauled, the power supply to the crane should be cut off completely, and all kinds of circuit breakers should be restored to the disconnected position (including the user's crane main circuit breaker). Sequence of power-off: Control handle is put to "zero position"; after each mechanism stops running, press the stop button of the linkage console to shut down the power, and disconnect the F1 air switch. If necessary, disconnect all air switches from translation and PLC cabinets and the power supply for weighing device.

2.5.2 临时停车：
2.5.2 Temporary Stop:

如起重机需要较长时间停止运行（如交接班），为防止非操作人员误操作，应断开主接触器，方法如下：在控制手柄零位，各机构停止运行后，按动联动台停止按钮失电使起重机的动力电源断开，其他电源没有必要可不断开。再次起动之前“手柄必须回到零位”，否则 PLC 报警并且没有运行指令，各机构不能工作。
If the crane needs to stop running for a longer period of time (such as handover), in order to prevent operation mistakes by non-operators, the main contactor should be disconnected as follows: after the control handle is in the zero position and mechanism stops running, press the stop button of linkage console to shut down the crane power supply, and other power supply can stay connected if necessary. Before starting again, "the handle must return to the zero position", otherwise the PLC will alarm; without operation command, the mechanism cannot work.

2.5.3 紧急停车：
2.5.3 Emergency Stop:

起重机发生带有危险性故障，如控制手柄、限位开关（尤其是起升机构上极限）失灵，电动机超速运行等现象，为防止事故或减小事故，需要紧急切断动力、控制电源时，按联动台上的红色蘑菇头式“总电源急停”按钮（此按钮按动后只有顺时针转动才能恢复），脱扣断开。起重机各机构的动力电源、控制电源断开，制动器断电制动。正常停车禁止使用此按钮。
If crane runs with dangerous faults, such as control handle and limit switch failure (especially the upper extreme limit of the lifting mechanism), motor overspeed and other phenomena, in order to prevent or mitigate accidents, the power should be off in emergency; when controlling power, press the red-mushroom-head-style main power emergency stop" button of linkage console (the button can only be restored by turning clockwise after it is pressed), and the release button will be disconnected. The power supply and control power supply of the crane mechanisms are shut off, and the brakes are on with powering off. This button is prohibited for normal stop.

2.6 紧急停车后的开车
2.6 Start-up after Emergency Stop

2.6.1 紧急停车按钮按动后只有顺时针转动才难恢复。
2.6.1 The emergency stop button is only difficult to recover by turning it clockwise after pressed.

2.6.2 排除故障时如断开其它电源，应按上述“开车”过程重新合闸、起动。如排除故障时没有断开其他空开和保护开关，只需在电源柜内将 F1 空开的手柄先下拉挂住触点机构，再上搬手柄空开才能合闸，而后按“起动”过程进行总电源接触器及其各控制柜的电源接触器的起动，起重机即可正常工作。
2.6.2 If other power supplies are off during troubleshooting, the switch should be closed and restarted according to the above "start-up" process. If troubleshooting is going on with the other air switches and protection switches on, one should only pull down the F1 air switch handle in the power cabinet to hitch on the contact mechanism, and pull it up to close, and start the main power contactor and power contactors in various control cabinets according to the "start" process, then the crane can work normally.

2.7 操作注意事项
2.7 Operation Precautions

2.7.1 抓斗下降运行中，离垃圾料位 1~2 米左右时手动降至一档低速运行，防止抓斗落到物料面时钢丝绳松绳过多，导致抓斗大幅度倾斜，钢丝绳和电缆线脱槽。因为抓斗上升或下降是通过变频器控制，变频器内设置好相应的加减速时间，在运行过程中通过联动台给定档位，从一档到四挡或从四挡到一档对应了变频器的加减速时间。比如下降过程中抓斗已经碰到物料面了，哪怕这个时候联动台从四挡直接回复到零位，变频器是还在持续有输出，抱闸不会抱死，只有当加减速时间到达之后，变频器才会停止输出，抱闸闭合，电机停转。这个时候钢丝绳已经松绳过多！
2.7.1 During the grab's descending, manual downshift to the first gear at a low speed is needed when it is about 1-2 meters above from waste level in order to prevent the wire rope from loosening excessively leading to a large tilt of the grab and the wire rope and the cable being out of the groove when the grab falls down to the material surface. Because the ascending or descending of the grab is controlled by the converter in which the corresponding acceleration and deceleration time is set, the gear position is set by the linkage console during operation, and the acceleration and deceleration time of the converter corresponds to that from the first gear to the fourth gear or from the fourth gear to the first gear. For example, in the descending process, the grab has touched the material surface, even if the linkage console restores zero position from the fourth gear and the converter is still outputting, the brake will not be locked. Only when the acceleration and deceleration time reach a certain level, the converter will stop outputting, the brake closes and the motor stops rotating. The wire rope has been loosened too much at this moment!

2.7.2 抓斗在抓完料开始上升时，特别是在抓斗大幅度倾斜或钢丝绳下降较多松绳时上升，先慢速运行提升让钢丝绳均匀受力，之后在开始加速，防止钢丝绳因快速起升产生的剧烈晃动与电缆线打架缠绕，引起电缆线拉断。
2.7.2 When the grab starts to ascend after grabbing the material, especially when the grab is greatly tilted or the wire rope drops many loosening ropes, slow lifting is needed first to enable the wire rope to bear evenly, then starting acceleration follows to prevent the wire rope from entangling with cable due to violent shaking caused by rapid lifting, resulting in broken cable.

2.7.3 小车往料口中心运行时，要及时减速防止小车速度过快导致抓斗剧烈碰撞坑壁。
2.7.3 When the trolley is running toward the center of the feed port, it needs to slow down in time to prevent the trolley from excessively fast speed leading to violent collision of the grab's pit wall.

2.7.4 称重运算的条件：
2.7.4 Conditions for weighing operation:

方案：第一，抓斗到达上限位位置。第二，小车到达前限位。第三，大车到料口中心，操作台的料口 1 指示灯或料口 2 指示灯亮起。第四，抓斗打开，满足上述四个条件开始记重。每次投料仅记重一次，不会重复记重，记重复位是抓斗下降至起升上减速位置以下自动进行复位。
Plan: First, the grab reaches the upper limit position. Second, the trolley reaches the

front limit. Third, when the crane reaches the center of the feed port, indicator 1 or 2 of operation console is on. Fourth, switch on the grab and start to weigh when the above four conditions are met. The weight is recorded only once for each feeding and will not be repeated. The repeated position is automatically reset when the grab descends above the lifting position and below deceleration position.

方案 2：在触摸屏上有料口记重按钮，只有在抓斗到达起升减速位置以上时，记重按钮才起作用，是为了防止倒料是误记重。每次投料仅记重一次，不会重复记重，记重复位是抓斗下降至起升上减速位置以下自动进行复位。
Plan 2: There is a weight-recording button on the touch screen, only when the grab reaches the lifting deceleration position or above, the button works, aiming to prevent the weight of material discharge from being mistakenly recorded. The weight is recorded only once for each feeding and will not be repeated. The repeated position is automatically reset when the grab descends above the lifting position and below deceleration position.

方案一和方案二相互兼容，但只执行一次称重记录。
Plan 1 and Plan 2 are compatible with each other, but only one weight recording is performed.

2.7.5 在起重机运行过程中，称重传感器处于晃动状态，此状态下传感器计量数据存在波动，记重之前保持设备不大幅度的晃动，数据记重比较精准。可以根据电脑上的当前重量数值变化判断是否已经稳定。
2.7.5 In the process of crane operation, the weighing sensor is in a shaking state, and measurement data fluctuate in this state. keeping the equipment away from violent shaking before weight recording will make the data fairly accurate. One can tell if it has been stabilized through the change in the current weight value on the computer.

2.7.6 为了防止抓斗溜钩和因操作不当引起的报警，变频器参数进行了防溜钩设置，当每次起升停稳之后再运行，有 1~3 秒时间的间隔，也就是说起升完全停稳之后在 1~3 秒内无法再次运行，第一是为了防止变频器刚停止之后突然换向导致变频器报警而引起抓斗溜钩。第二是为了防止电机刚启动的瞬间电机力矩不够导致溜钩。
2.7.6 In order to prevent the grab from hook slipping and alarm caused by improper operation, the converter parameters are set to prevent hook slipping. When operating after lifting stops each time, there is an interval of 1-3 seconds, which means that the lifting cannot be operated again for 1-3 seconds after it completely stops. First is to prevent a sudden reversal of the direction after the converter stops, resulting in an alarm of the converter, which may cause the grab hook slipping. The second is to prevent the motor from slipping the hook due to insufficient motor torque at the moment when the motor is first started.

2.7.7 起升变频器每次由静止到启动运行时有 1.5 秒的电机励磁时间，制动器不会马上打开，电机也不会马上运行，是为了让电机运行时有足够的启动转矩。
2.7.7 There is a 1.5-second motor excitation time when the lifting converter runs from standstill to startup each time. The brake will not operate immediately and so as the motor, so that the motor can run with enough starting torque.

2.7.8 当称重软件发生卡机或死机，请重新启动软件。
2.7.8 When the weighing software is stuck or halted, please restart.

2.8 起升机构保护机制
2.8 Lifting Protection Mechanism

2.8.1 起升抱闸空开 F12：
2.8.1 Lifting Brake Air Switch F12:

当起升抱闸电机故障或抱闸电机电源线因断裂，破皮引起的接地短路，空开跳闸，起升机构会停止运行。
When the lifting brake motor fails or the brake motor power line is short circuited to ground due to broken or torn wire, the air switch trips and the lifting mechanism will stop running.

2.8.2 起升风机空开 F13：
2.8.2 Lifting Fan Air Switch F13:

当起升风机电机故障或抱闸电机电源线因断裂，破皮引起的接地短路，空开跳闸，起升机构会停止运行。
When the lifting fan motor fails or the brake motor power line is short-circuited to ground due to broken or torn wire, the air switch trips and the lifting mechanism will stop running.

2.8.3 起升下限位 K2：
2.8.3 Lower Lifting Limit Position K2:

当抓斗在闭合状态下到达设定的下降位置后，起升机构不能继续向下运行。
When the grab reaches the set lowering position in the closed state, the lifting mechanism cannot continue downward.

2.8.4 起升上减速 K3：
2.8.4 Lifting Upper Deceleration K3:

起升减速位置设置在距离料口 1~2 米左右的水平面位置，到达该位置后起升机构会自动切除高速档位。
The lifting deceleration position is set at a horizontal position about 1-2 meters away from the feed port, and the lifting mechanism will automatically cut off the high-speed gear after reaching this position.

2.8.5 起升上限位 K4：
2.8.5 Lifting Upper Limit Position K4:

起升停止位置设置在距离料口上方 1~2 米左右的高度，到达该位置起升无法继续向上运行。
The lifting stop position is set at a height of about 1-2 meters above the feed port, and the lift cannot continue to run upwards when it reaches this position.

2.8.6 如果起升机构只能单方向运行，请检查限位。
2.8.6 If the lifting mechanism can only operate in one direction, check the limits.

2.8.7 起升变频器报警：详细信息根据变频器面板报警信息。
2.8.7 Lifting converter alarm: see alarm message on the converter panel for details.

2.9 大车机构保护机制
2.9 Crane Protection Mechanism

2.9.1 大车抱闸空开 F15：
2.9.1 Crane Brake Air Switch F15:

当大车抱闸电机故障或电机电源线因断裂，破皮引起的接地短路，空开跳闸，大车机构会停止运行。，
When the crane brake fails or motor power line is short-circuited to ground due to broken or torn wire, the air switch trips and the crane mechanism will stop running.

2.9.2 大车防撞限位 K8：
2.9.2 Crane Crash Limit K8:

红外开关发射距离为 3 米，当两车靠近时，距离接近 3 米时大车无法继续运行。
The infrared switch transmits at a distance of 3 meters, and when the two vehicles get close to each other, the crane cannot continue to run when the distance is close to 3 meters.

2.9.3 大车减速限位 K7，停止限位 K6：
2.9.3 Crane Deceleration Limit K7, Stop Limit K6:

大车的限位装置采用的是电感式传感器，在大车端梁侧部焊接有感应铁，当感应铁靠近传感器时，传感器接通继电器，使限位断开，在特定的地点进行减速和停止。
The limit device of the crane is an inductive sensor with a sensing iron welded to the side of the end beam. When the sensing iron is close to the sensor, the sensor connects the relay to disconnect the limit, decelerating and stopping at a specific place.

2.9.4 如果只能往单方向运行，请检查限位。
2.9.4 If only one-direction operation is allowed, please check the limits.

2.9.5 大车变频器报警：详细信息根据变频器面板报警信息。
2.9.3 Crane converter alarm: see alarm message on the converter panel for details.

2.10 小车机构保护机制
2.10 Trolley Protection Mechanism

2.10.1 小车抱闸空开 F17：
2.10.1 Trolley Brake Air Switch F17:

当小车抱闸电机故障或电机电源线因断裂，破皮引起的接地短路，空开跳闸，小车机构会停止运行。，
When the trolley brake fails or motor power line is short-circuited to ground due to broken or torn wire, the air switch trips and the trolley mechanism will stop running.

2.10.2 小车前减速 K14，小车前限位 K15，小车后减速 K13，小车后限位 K12
2.10.2 Trolley front deceleration K14, trolley front limit K15, trolley rear deceleration K13, trolley rear limit K12

小车的限位装置采用的是电感式传感器，在小车侧边焊接有感应铁，轨道旁边焊

接有传感器支架，当感应铁靠近传感器时，传感器接通继电器，使限位断开，在特定的地点进行减速和停止。
The limit device of the trolley is an inductive sensor with a sensing iron welded to the side of the trolley and a sensor bracket welded to the side of the track. When the sensing iron is close to the sensor, the sensor connects a relay to disconnect the limit, decelerating and stopping the trolley at a specific place.

2.10.3 如果只能往单方向运行，请检查限位。
2.10.3 If only one-direction operation is allowed, please check the limits.

2.10.4 大车变频器报警：详细信息根据变频器面板报警信息。
2.9.3 Crane converter alarm: see alarm message on the converter panel for details.

2.11 抓斗机构保护机制
2.11 Grab Protection Mechanism

2.11.1 抓斗空开 F18，当抓斗电机故障或电机电源线因断裂，破皮引起的接地短路，空开跳闸，抓斗机构会停止运行。
2.11.1 Grab air switch F18, when the grab motor fails or motor power line is short-circuited to ground due to broken or torn wire, the air switch trips and the grab mechanism will stop running.

2.11..2 更换抓斗电缆时请注意控制线和电源线的接线顺序，确保电机相序和电磁阀方向准确。
2.11..2 When replacing the grab cable, please pay attention to the wiring sequence of the control and power wires to ensure that the motor phase sequence and solenoid valve direction are accurate.

2.12 电源保护机制
2.12 Power Supply Protection Mechanism

2.12.1 相序继电器 K1：
2.12.1 Phase Sequence Relay K1:

检测三相电源的过压值，欠压值，相序，在继电器上有相应的故障指示灯，如果出现不能启动电源请检查。
Check the over-voltage value, under-voltage value and phase sequence of the three-phase power supply. There is a corresponding fault indicator on the relay, please check if the power supply cannot be on.

2.12.2 遥控与联动台相互转换时请在联动台上旋转选择开关，保障在各自操作状态下机构能正常启动运行。选择遥控操作时只能作为检修状态使用，正常情况下使用联动台操作。
2.12.2 When the remote control and the linkage console are switched to each other, please rotate the selector switch on the linkage console to ensure that the mechanism can start and run normally under their respective operating conditions. When remote control operation is selected, it can only be used in an overhaul state, and the linkage console is used for operation under normal conditions.

2.12.3 启动前请确认遥控器或联动台的急停开关已处于打开状态。
2.12.3 Before startup, please make sure the emergency stop switch on the remote control or linkage console is on.

2.13 故障
2.13 Malfunctions

2.13.1 当某一机构无法运行时，请在触摸屏或控制柜内检查保护开关的状态和变频器的报警代码，可进行手动恢复和复位。
2.13.1 When a mechanism cannot be operated, please check the status of the protection switch and the alarm code of the converter on the touch screen or in the control cabinet, which can be manually restored and reset.

2.13.2 当某一个机构无法运行时，先参考触摸屏上的故障显示以及各个机构运行要满足的运行条件。
2.13.2 When a mechanism cannot be operated, please first refer to the fault display on the touch screen and the operating conditions to be met for each mechanism to operate.

第三章 维护与保养
Chapter III Maintenance and Servicing

起重机的正确使用与维护保养对安全生产和起重机的使用寿命有很大关系，因此，必须对起重机进行定期检查和检测，检查或检测必须符合 GB6067-2010《起重机械安全规程》和其它有关文件规定的要求。日常维护与保养可按下述内容进行。
The correct use and maintenance of cranes has a great relationship with the safety production and the service life of cranes. Therefore, cranes must be regularly inspected and tested, and the inspection or test must be in line with the GB6067-2010 *Safety Rules for Lifting Appliances* and other relevant requirements stipulated in the documents. Routine maintenance and servicing can be performed as follows.

管理工作是至关重要的，按照现代管理理念，使工作人员有明确的责、权、利。应建立起重机的技术档案，记录应包括交接班时间；设备运行状况是否正常等。操作人员每班应对起重机进行检查，做好交接班。有问题时应作事故记录：1.事故发生时间：年、月、日、时、分；2.故障现象简述：出现在什么机构（部位），有否光、火、烟、味等；3.故障程度：是否造成停产，停产时间段；4.发生故障的原因，在操作上是如何进行的；5.采取的措施、检修方法及实际检修的问题等。所有记录当班人在签字（表格见附件）。
Management is essential to provide staff with clear responsibilities, rights and benefits in accordance with modern management concepts. Technical files of cranes should be established which should record shift handover time and whether the equipment is in normal operating condition. The operator should check the crane in every shift and ensure good shift handover. When there is a problem, an accident record should be made: 1. The time of the accident: in which year, month, day, hour, minute; 2. A brief description of the failure: in what institutions (parts), whether there is light, fire, smoke, smell, etc.; 3. The extent of the failure: whether it caused the shutdown of production, the period of time; 4. The reasons for the failure, the process of operation; 5. The measures, the overhaul method and the actual problems and so on. All signatures of persons on duty are recorded (see the table in the attachment).

3.1 起升机构的维护与保养
3.1 Maintenance and Servicing of Lifting Mechanism

3.1.1 电机在正常运行时的温升不应超过样本规定的温度。用温度计测量，铁芯或机壳温升一般不超过 60K。轴承温升一般不超过 950℃。检修人员应每天进行一次温度探查，每周进行一次温度测量。
3.1.1 The temperature rise of the motor during normal operation should not exceed the temperature specified in the sample. Measured with a thermometer, the temperature rise of the iron core or case should not exceed 60K. The temperature rise of bearings should not exceed 950℃. Overhaul staff should check temperature daily and measure temperature weekly.

3.1.2 注意电机的气味、振动和噪声，在闻到焦味、发现不正常的振动或其它杂音，应立即停机检查。
3.1.2 Pay attention to the odor, vibration and noise of the motor. When burnt odor is smelt, abnormal vibration or other noise is found, it should be stopped immediately

for inspection.

3.1.3 定期检查轴承并补充 2 号锂基润滑脂。
3.1.3 Regularly check the bearings and replenish lithium grease No. 2

3.1.4 注意保持电机的清洁，不允许有水滴、油污及杂物进入电机内部。
3.1.4 Keep the motor clean and do not allow water droplets, oil and debris to enter the inside of the motor.

3.1.5 变频电机因变频器高次谐波的影响，电机噪声和振动比电网供电大，特别是谐振频率一致而发生共振时。可采用增强系统刚度或利用变频器的频率跳变功能，避开谐振频率，实现平滑运转。
3.1.5 Due to the influence of high harmonics of the converter, noise and vibration of the motor is greater than that of the grid supply, especially when the resonant frequency is consistent and resonance occurs. Adopting system stiffness or using frequency hopping function of the converter can be used to avoid resonant frequency and realize smooth operation.

3.1.6 其它内容详见《电机使用维护说明书》。
3.1.6 For other contents, please refer to *Motor Use and Maintenance Manual*.

3.1.7 电动机定子绝缘电阻小于 0.5MΩ，必须对电动机进行干燥处理。
3.1.7 When motor stator insulation resistance is less than 0.5MΩ, the motor must be dried.

3.2 起升减速器的维护与保养
3.2 Maintenance and Servicing of Lifting Reducers

3.2.1 减速器润滑油应选用 GB5903 中的 N68、N100、N150、N220 中负荷极压齿轮油，南方宜选用高牌号，北方宜选低牌号。
3.2.1 Lubricating oil of reducer should select medium load extreme pressure gear oil of N68, N100, N150, N220 of GB5903, and the South is appropriate to choose a high grade, while the North is appropriate to choose a low grade.

3.2.1 首次使用 3 个月后应换油一次，以后连续使用一般六个月换油一次，间断使用可一年换油一次。
3.2.1 The oil should be changed once after 3 months of first use, and then once every six months for continuous use, and once every year for intermittent use.

3.2.3 减速器要经常检查油位。注意：任何时候，油位都不能低于油标下限。检查应在减速器停止运转并充分冷却后进行。
3.2.3 The oil level of the reducer should be checked frequently. Note: at no time should the oil level fall below the lower limit of the oil leveler. The inspection should be carried out after the gearbox has stopped running and has cooled down sufficiently.

3.2.4 减速器在使用过程中，各联结件、紧固件不得松动。
3.2.4 During the use of the reducer, junction piece and fasteners shall not be loosened.

3.2.5 使用中如发现结合面漏油，请拆开渗漏处零件涂 601 密封胶；如发现油封漏油，请按原油封型号更换。
3.2.5 If oil leakage is found on the joint surface during use, please disassemble the

leaking parts and apply 601 sealant; if oil leakage is found on the oil seal, please replace it according to the original oil seal type.

3.2.6 其它内容详见《起重机减速器使用说明书》。
3.2.6 For other contents, please refer to *Use Manual of Crane Reducer*.

3.3 起升制动器的维护与保养
3.3 Maintenance and Servicing of Lifting Brakes

3.3.1 推动器是否有漏油和渗油现象。油液是否足量，不足时应入适当的油，油应定期更换。
3.3.1 If the propeller has the problem of oil leakage and seeping. Whether the oil is sufficient, appropriate oil should be added when in an insufficient condition, and oil should be replaced periodically.

3.3.2 电机引入线的绝缘是否良好。
3.3.2 If the insulation of the motor's lead-in wires is good.

3.3.3 制动系统各部分应准确灵活的动作，应固定的螺母、保险垫片等不应松动。
3.3.3 All parts of the braking system should act accurately and flexibly, and the nuts, safety shims, etc. that should be fixed should not be loose.

3.3.4 制动弹簧、制动臂、底座、杠杆、螺杆等如出现裂纹应及时更换。
3.3.4 Brake springs, brake arms, bases, levers, screws, etc. should be replaced if cracked.

3.3.5 制动弹簧出现塑性变形应更换。
3.3.5 Brake springs should be replaced if they show plastic deformation.

3.3.6 制动瓦应正确地贴合在制动轮上，其实际接触面积不应小于理论接触面积的 50%。
3.3.6 Brake tiles shall fit correctly on the brake wheel and their actual contact area shall not be less than 50% of the theoretical contact area.

3.3.7 制动器主要绞接处均有注油油杯，每月应检查注油。
3.3.7 The main strand joints of the brake are oiled with oil cups, which should be checked and oiled every month.

3.3.8 在抱闸情况下，应保证推动器活塞杆伸出 10mm 左右。
3.3.8 In the case of contracting brake, it should be ensured that the propeller's piston rod extends about 10mm.

3.3.9 经常检查和调整制动弹簧的长度，以保证正确的制动力矩。
3.3.9 Check and adjust the length of the brake spring frequently to ensure the correct braking torque.

3.3.10 当制动衬垫的厚度磨损到下表中的数值时，应更换制动衬垫。
3.3.10 When the thickness of the brake liner wears to the values in the table below, the brake liner should be replaced.

制动轮直径 mm Brake wheel	200~250	315~400	500~630	710~800

diameter mm				
衬垫厚度 mm Thickness of liner mm	4	5	6	7.5

3.3.11 制动轮沾有油污时，须用煤油清洗制动轮表面和制动垫垫。
3.3.11 When the brake wheel is contaminated with oil, the surface of the brake wheel and the brake pad shall be cleaned with kerosene.

3.3.2 其它内容详见《制动器使用说明书》
3.3.2 For other contents, please refer to *Use Manual for Brakes*.

3.4 卷筒联轴器的维护与保养
3.4 Maintenance and Servicing of Drum Couplings

3.4.1 卷筒联轴器用 2 号锂基脂润滑，在高温条件下宜用 3 号锂基脂。每月应注油一次，到油脂溢出为止。
3.4.1 Drum couplings are lubricated with No. 2 lithium grease, and No. 3 lithium grease is recommended under high temperature conditions. It should be oiled once a month until the grease overflows.

3.4.2 减速器出轴端设有挡板和紧固螺栓，联轴器和卷筒连接螺栓须经常检查其紧固性。
3.4.2 The shaft end of the reducer is equipped with a baffle plate and fastening bolts, and the coupling and drum connecting bolts shall be frequently checked for tightness.

3.4.3 其它内容详见《WZL 卷筒联轴器使用说明书》。
3.4.3 For other contents, please refer to *Use Manual of WZL Drum Coupling*.

3.5 钢丝绳的维护与保养
3.5 Maintenance and Servicing of Wire Ropes

3.5.1 为防止钢丝绳松懈和降低其强度，抓斗工作时，应避免钢丝绳打转扭结。
3.5.1 In order to prevent the wire rope from slackening and reduce its strength, the wire rope shall be avoided from spinning and kinking when the grab is in operation.

3.5.2 钢丝绳应定期润滑。润滑前应用浸有煤油的抹布清洗旧油。绝对禁止用金属刷或其它尖锐器具清洗钢丝绳上的污物；禁止使用酸性或其它具有腐蚀性的润滑剂。
3.5.2 Wire ropes should be lubricated regularly. A kerosene-soaked rag should be used to clean the old oil before lubrication. Absolutely prohibit the use of metal brushes or other sharp instruments to clean the dirt on the wire rope; prohibit the use of acidic or other corrosive lubricants.

3.5.3 钢丝绳磨损后的报废按国家有关标准执行。
3.5.3 The scrapping of wire rope after wearing is carried out according to the relevant state standard.

3.5.4 更换或窜动钢丝绳后应重新调整极限上限位开关。钢丝绳实际安装长度与

设计长度一致,更换钢丝绳时保证新旧钢丝绳等长和绳两端固定相同及卷筒未转动，则不需调整起升高度限位器。
3.5.4 The extreme upper limit switch should be readjusted after replacing or tampering with the wire rope. The actual installation length of the wire rope is consistent with the design length, when replacing the wire rope, make sure that the new and old wire ropes are of equal length and the two ends of the rope are fixed in the same way and the drum is not rotated, then there is no need to adjust the lifting height limit switch.

3.6 抓斗的维护与保养
3.6 Maintenance and Servicing of Grab

1.6.1 首次起动工作约 50 小时后，清洁液压过滤器。
1.6.1 Clean the hydraulic filter after approximately 50 hours of initial start-up operation.

1.6.2 首次使用 500 个工作小时应换油一次，以后每使用 10000 个工作小时换油一次。
1.6.2 The oil should be changed once for 500 working hours of first use and once for every 10,000 working hours thereafter.

1.6.3 定期检查：油位；插头的连接和张紧释放装置；软管和螺纹接头与油缸的密封性；钢丝绳与抓斗的连接；系统油压（开启：120bar，闭合 190bar）。
1.6.3 Periodic inspection: oil level; plug connections and tension release devices; sealing of hoses and threaded joints to cylinders; connection of wire rope and grab; system oil pressure (open: 120bar, closure 190bar).

1.6.4 当抓斗停止工作时，应切断电源，使活塞杆回收，保证抓斗处于打开状态。
1.6.4 When the grab stops working, the power supply should be cut off so that the piston rod can be recovered to ensure that the grab is in an open state.

1.6.5 每周用 2#锂基润滑脂润滑一次所有润滑点。
1.6.5 Lubricate all lubrication points once a week with 2# lithium grease.

3.7 称量装置的维护与保养
3.7 Maintenance and Servicing of Weighing Devices

称重装置由安装在小车架下方车轮轴承座上方的四只托利多传感器组成，称量控制系统（包含监控计算机，打印机组成）安装在垃圾吊的控制室内。
The weighing device consists of four Toledo sensors mounted above the wheel bearing seat below the trolley frame, and the weighing control system (consisting of a monitoring computer and printer) is installed in the control room of the waste crane.

3.7.1 称量装置调整完毕，非维护人员不得随意调整任何部件。
3.7.1 After the weighing device has been adjusted, non-maintenance personnel shall not be allowed to adjust any parts.

3.7.2 每个季度检查传感器零点和报警变化情况，发现问题及时调整，以消除元件漂移带来的误差。

3.7.2 Check the sensor zero point and alarm changes every quarter and make timely adjustments when problems are found to eliminate errors caused by component drift.

3.7.3 载荷限制器是安全保护装置，不可因有限制器而忽视安全操作规程。
3.7.3 The load limiter is a safety protection device, which should not be ignored because of its possession of the limiter.

3.7.4 未经许可，禁止进入“标定”状态，调整标定值。
3.7.4 It is forbidden to enter the "calibration" state and adjust the calibration value without authorization.

3.7.5 其它内容详见称重传感器《使用说明书》等相关资料。
3.7.5 For other contents, please refer to the *Use Manual* of the weighing sensor and other related materials.

3.8 卷筒的维护与保养
3.8 Maintenance and Servicing of Drums

本项目的垃圾吊选用同步一体式电缆卷筒，供电电压 380V；50Hz。
The waste crane of this project uses synchronized one-piece cable drum with a supply voltage of 380V; 50Hz.

3.8.1 如果滑环出现电蚀、烧伤，应用细砂布磨光或机加工抛光，否则，需更换集电环，电刷的更换视其和集电环的接触情况而定。
3.8.1 If the slip ring shows galvanic corrosion or burns, it should be polished with a fine emery cloth or machine polished, otherwise, it is necessary to replace the collector ring, and the replacement of collector rings depends on the contact between it and the collector ring.

3.8.2 当收卷电缆出现电缆过松或释放电缆出现电缆过紧时，应立即停机，拆开压板螺栓，调整电缆的松紧，上、下旋转电缆卷筒，调整好电缆的松紧适度后，再恢复拧紧压板螺栓。
3.8.2 In case of loose reel-up or tight reel-out, the machine should be stopped immediately, disassemble the pressure plate bolt, adjust the tightness of the cable, rotate the cable drum up and down, adjust the cable's looseness and tightness appropriately, and then resume tightening the follower bolt.

3.8.3 当电缆卷筒不能正常收放电缆时，应检查钢丝绳卷筒是否能够正常转动或传动机械部分是否卡住或磨损。
3.8.3 When the cable drum cannot reel up and out the cable normally, it should be checked whether the wire rope drum can rotate normally or whether the transmission mechanical part is jammed or worn.

3.9 大、小车运行机构的维护与保养
3.9 Maintenance and Servicing of Crane and Trolley Running Mechanisms

3.9.1 大车运行机构采用分别驱动，三合一的制动器必须调整到保证起重机两端同步起动和停止。

3.9.1 The crane running mechanism is driven separately and the three-in-one brake must be adjusted to ensure synchronized start and stop at both ends of the crane.

3.9.2 车轮轴承座应保持良好的润滑状态，注油量应为轴承体空腔的 2/3。
3.9.2 Wheel bearing seats should be kept well lubricated, and the amount of oil injection should be 2/3 of the cavity of the bearing body.

3.9.3 当发现轴承温度过高或噪声过大时，必须停车检查。有缺陷的轴承应立即更换。
3.9.3 When the bearing temperature is found to be too high or the noise is too loud, it must be stopped and inspected. Defective bearings should be replaced immediately.

3.9.4 大、小车的从动车轮轴上装有位置编码器，应定期检查其运转情况。
3.9.4 Position encoders are fitted to the axles of the driven wheels of crane and trolley, and their operation should be checked regularly.

3.9.5“三合一”减速器的制动力矩对于相应的型号其大小是固定不变的，出厂前已调好，并用火漆标记。客户不能自行调整。
3.9.5 The braking torque of the "three-in-one" reducer is fixed for the corresponding model and is adjusted before delivery and marked with fire paint. Customers cannot make their own adjustments.

3.9.6 定期检查“三合一”减速器的制动气隙，并根据型号规格调整至 0.2~0.5mm。
3.9.6 Regularly check the brake air gap of the "three-in-one" reducer and adjust it to 0.2~0.5mm according to the model specification.

3.9.7 定期检查车轮，发现裂缝、压痕及过渡磨损应及时参照 GB6067-2010《起重机安全规程》进行检修或更换。
3.9.7 Regularly inspect wheels, when cracks, indentations and transitional wear are found, they should be promptly overhauled or replaced with reference to GB6067-2010 *Safety Rules for Lifting Appliances*.

3.10 金属结构（桥架、小车架）的维护与保养
3.10 Maintenance and Servicing of Metal Structures (Bridges, Trolley Frames)

3.10.1 金属结构的检查
3.10.1 Inspection of metallic structures

桥式起重机的金属结构每年检查 1～2 次，重点为连接的松动、脱落、结构材料和焊缝的裂纹开裂、桥架变形、结构件的腐蚀。金属结构的检查内容和判定标准见表 3-10-1 所示。
The metal structure of the bridge crane is inspected 1 to 2 times a year, focusing on loosening and dislodging of connections, cracking and crazing of structural materials and welds, deformation of the bridge, and corrosion of structural components. The inspection contents and determination criteria for metal structures are shown in Table 3-10-1.

表 3-10-1 金属结构检查内容与判定标准
Table 3-10-1 Metal Structure Inspection Content and Determination Criteria

检查项目 Inspection Items		检查内容 Inspection Contents	判定标准 Determination Standard
主梁 Main Beam	主梁变形 Main Beam Deformation	检测主梁在起吊额定载荷时跨中的挠度，以及水平旁弯和其他变形。 Detect deflections in the span of the main beam when lifting the rated load, as well as horizontal sidewise bending and other deformations.	下挠应＜S/1000；旁弯及变形值应符合规定标准 Deflection should be <S/1000; sidewise bending and deformation values should be in line with specified standards
	结构件 Structural Parts	检查金属结构件有无裂纹、腐蚀、异常变形、整体扭曲、局部失稳 Check if metal structural components have cracks, corrosion, abnormal deformation, overall distortion, and local instability 检查连接部分有无松动，脱落、裂纹、腐蚀 Check if the connecting parts have looseness, detachment, cracks and corrosion.	不得有裂纹、明显腐蚀、异常变形、明显扭曲和局部失稳 There shall be no cracks, obvious corrosion, abnormal deformation, obvious distortion and localized instability. 不得有松动、脱落、裂纹、腐蚀 There shall be no looseness, detachment, cracks and corrosion
	其他 Others	检查金属结构表面防护 Check surface protection of metal structures	不得有油漆起泡、剥落、明显锈蚀 There shall be no lacquer bubble, flake, and visible rust
小车 Trolley	结构件 Structural Parts	检查有无裂纹、变形、开裂 Check for cracks, deformation, and splitting 检查钢结构表面防护 Check surface protection of steel structure 检查各连接有无松动、脱落 Check if all connections have looseness and detachment.	无裂纹、变形、开裂 No cracks, deformation, and cracking 不得有油漆起泡、剥落、明显锈蚀 There shall be no lacquer bubble, flake, and visible rust 无松动、脱落 No looseness or detachment
主梁连接 Main Beam Connection		检查连接处母材及焊缝区有无裂纹 Check if the base metal and weld area of the joint have cracks 检查螺栓等是否紧固可靠 Check bolts, etc. for tightness and reliability	无裂纹 No crack 应紧固可靠 They should be tightness and reliability

3.10.2 机构的检查与维护
3.10.2 Inspection and maintenance of mechanisms

A）起升机构的检查，起升机构检查的项目内容见表 3-10-1、表 3-10-2、表 3-10-4，对于起升机构和运行机构相同的零部件、设备等，可参照表 3-10-2、表 3-10-3、表 3-10-4 相应的项目内容及判定标准进行。根据小车式电动葫芦各部分结构在安全运转上的重要程度，使用频繁程度以及是否易损件等来确定检查周期，一般分为三个等级：I 级，每月必须检查一次；II 级，每三个月必须检查一次；III级，每六个月必须检查一次。

A) Inspection of the lifting mechanism, the item content of the lifting mechanism inspection is shown in Table 3-10-1, Table 3-10-2, and Table 3-10-4, for the same parts, equipment, etc. of the lifting mechanism and the operating mechanism, you can refer to the corresponding item content and determination criteria of Table 3-10-2, Table 3-10-3, and Table 3-10-4. Determine the inspection cycle based on the importance, the degree of frequent use, and whether being wearing parts of each part's structures of the trolley-type electric hoist in the safe operation, which generally divided into three levels: level I, must be inspected once a month; level II, must be inspected every three months; level III, must be inspected once every six months.

表 3-10-2 小车日常检查项目及其要求

Table 3-10-2 Trolley Daily Inspection Items and Requirements

检查项目 Inspection Items	要求 Requirements
作业地点 Operation site	在操作者步行范围内无障碍物 No obstacles within walking distance of the operator
遥控按钮装置 Remote control button unit	起升、下降、左右运行动作应灵敏，准确；同时按动一组按钮不得动作 Lifting, lowering, left and right operations should be sensitive and accurate; press a group of buttons simultaneously does not realize lifting, lowering, left and right operations simultaneously
限位器 Limiter	空载上升至极限位置时，限位开关应准确可靠 Limit switches should be accurate and reliable when no load is raised to the limit position
钢丝绳 Wire rope	按 GB5972—1986 中 2.4.1.1 进行日常观察 Daily observation is carried out according to 2.4.1.1 in GB5972-1986
制动器 Brake	起升、下降及运行的制动应灵敏可靠 The brakes for lifting, lowering and operation should be sensitive and reliable.
抓斗供电电缆 Grab power cables	动作正常，出绳通畅 Normally action, smoothly reeling out

表 3-10-3 小车月检查及其要求和等级

Table 3-10-3 Monthly Trolley Inspections and Requirements and Ratings

检查项目 Inspection Items		要求 Requirements	等级 Level
抓斗连接装置 Grab attachments	销轴 Pin shaft	不得有异常磨损， There shall be no abnormal wear,	I
	外观 Appearance	不得有损伤，挡轴板挡圈及销不得有松动，钩口闭锁装置应动作正常 There shall be no damage, no looseness of the retaining ring and pin of the shaft stop plate, and the hook locking device should operate normally.	I
	工作状态 Working status	回转应灵活，回转不得有损伤，装置连接牢固 Slew should be flexible and not be damaged, and the device is firmly connected	III
	墙板 Wall panel	连接螺栓不得有松动 Connecting bolts must not be loose	III
钢丝绳 Wire rope	绳端固定状况 Rope end	钢丝绳各尾端固定应牢固可靠，不得有异常 Each end of the wire rope should be fixed firmly and reliably, and there should be no abnormality.	I

	fixing condition		
	外观 Appearance	不得有扭结、灼伤及明显的松散、腐蚀等缺陷，绳上应有润滑油脂 There shall be no kinks, burns and obvious looseness, corrosion and other defects, there shall be lubricating grease on the rope.	I
	使用的程度（标准报废） Extent of use (Standard Scrapping)	按 GB5972 中 2.5.1—2.5.11 条的规定执行. Execute according to the provisions of clause 2.5.1-2.5.11 in GB5972.	
减速器 Reducer	齿轮润滑状况 Gear lubrication condition	齿轮箱体中应定期加润滑油 Regular lubricating oil should be added to the gear box	II
电缆 Cable	外观 Appearance	电缆不得有外伤，异常的弯曲或扭转、老化等缺陷 The cable shall not have defects such as trauma, abnormal bending or twisting, aging, etc.	II
大小车供电装置 Power supply unit for crane and trolley	装配状况 Assembly condition	保证工字钢安装的水平，高低，弯曲都在误差范围内 Ensure that the level, height and bending of the I-steel are within the tolerance range	III
	工作状况 Working condition	电缆小滑车运行平稳，顺畅，不卡阻。 The cable jigger runs smoothly and without jamming.	II

表 3-10-4 小车年检项目及要求

Table 3-10-4 Trolley Annual Inspection Items and Requirements

检查项目 Inspection Items		要求 Requirements
车轮 Wheel	轮缘 Rim	轮缘厚度的磨损量不得超过原厚度的 50%，轮缘与轨道的侧向总间隙应小于车轮踏面宽度的 50% Wear of the rim thickness shall not exceed 50% of the original thickness, and the total lateral clearance between the rim and the track shall be less than 50% of the tread width of the wheel.
	踏面 Tread	按踏面直径测量磨损量应小于原尺寸的 5%，踏面直径差应小于公称直径的 1%，圆度差应小于 0．8 According to the tread diameter to measure wear should be less than 5% of the original size, tread diameter difference should be less than 1% of the nominal diameter, roundness difference should be less than 0.8%
	外观 Appearance	不得有裂纹、损伤 There shall be no cracks, and damage
齿轮 Gear	起升机构齿轮磨损 Wear of lifting mechanism gears	第一级齿轮允许磨损量应小于原齿厚的 15%；其它应小于 20% Allowable wear of the first stage gear should be less than 15% of the original tooth thickness; the others should be less than 20%.
	运行机构齿轮磨损	第一级齿轮允许磨损量应小于原齿厚的15%；其它齿轮应小于25%，开式齿轮应小于 30%

	Wear of running mechanism gears	Allowable wear of first stage gears should be less than 15% of the original tooth thickness; other gears should be less than 25% and open gears should be less than 30%.
	齿面缺陷检查 Inspection of teeth surface defects	齿部不得有裂纹、断齿；齿面点蚀损坏不得大于啮合面的 30%，且深度不得超过齿厚的 10% The teeth shall not have damage and fracture; teeth surface pitting damage shall not be greater than 30% of the mesh, and the depth shall not exceed 10% of the tooth thickness.
制动器 Brake		按月检要求重复检查 Repeat inspection as required by monthly inspection
卷筒 Drum		不得有裂纹，筒壁磨损量小于原壁厚的 10%。 There shall be no cracks, and the wear of the cylinder wall shall be less than 10% of the original wall thickness.
键、花键 Key and spline		键与键槽连接不得有松动和变形以及异常磨损。 The key and keyway connection must not be loose and deformed or abnormally worn.
滚动轴承 Rolling bearing		不得有破损和裂纹。 There shall be no damage or cracks.
油封 Oil seal		配合面不准有裂纹。 No cracks are allowed on the mating surfaces.
电缆 Cable		按月检要求重复检查。 Repeat as required for monthly inspections.
限位开关 Limit switch		按月检要求重复检查。 Repeat as required for monthly inspections.
导电体与接地螺钉电阻 Conductor and grounding screw resistance		不得小于 0.1Ω。 Not less than 0.1Ω.
回路的对地绝缘电阻 Insulation resistance to ground of the circuit		不得小于 1.5Ω。 Not less than 1.5Ω.

B）起重机运行机构的检查维护：起重机和小车的运行机构的检查项目、内容及判定标准见表 3-10-5

B) Inspection and maintenance of crane running mechanism: inspection items, content and judgment standards of running mechanism of crane and trolley are shown in Table 3-10-5

表 3-10-5 起重机和小车运行机构检查项目及要求

Table 3-10-5 Inspection Items and Requirements of Crane and Trolley Operating Mechanism

检查项目 Inspection Items		检查内容 Inspection Contents	判定标准 Determination standard
电动机 Motor	安装支架 Mounting bracket	支架有无裂纹；连接有无松动、脱落，橡胶垫无老化 Bracket has no cracks; connection is not loose or spalling, and rubber gasket is not worn-out	无裂纹、无松动或脱落 No cracks, no loosening or spalling
减速器 Reducer	润滑油的检查 lubricating oil	最初使用 100 小时后，清净内部，换上新的润滑油，以后每隔 2500 小时	发现油有污染，则应换润滑油

	inspection	换油，检查油样包括颜色颜、浓度， After the initial use of 100 hours, clean the interior and set up the new lubricating oil. Then every 2,500 hours change the oil and check the oil samples, including the color and the concentration.	If the lubricating oil is found to be contaminated, it should be replaced.
	润滑油的更换 Lubricating oil replacement	在运行温度下换油， Change the oil under the running temperature	切断传动装置电源，等减速器冷却下来无燃烧危险为止 Cut off the power supply of the transmission device, wait for the speed reducer to cool down without the risk of combustion.
		将油全部排除，再通过油孔注入新的油 Remove all the oil, and then inject new one through the oil holes	到油位塞孔有油流出为止 until the oil flows out of the oil level plug holes.
轴 Shaft	转轴、心轴、传动轴 Rotary shafts, mandrels, transmission shafts	检查有无变形与磨损；传动轴是否有振摆；键及键槽松动、变形、裂纹 Check for deformation and wear; oscillation in transmission shaft; key groove loose, deformation, and cracks	无变形和明显磨损；无明显振摆；无松动、变形、裂纹 No deformation and obvious wear; no obvious oscillation; no loose, deformation, and cracks
轴承 Bearing	滚动轴承 Rolling bearing	检查有无裂纹与损伤；润滑状况 检查在空载和负载工况下有无异常振动、发热、噪声 Check for cracks and damage; lubrication condition Check for abnormal vibration, heat, noise in the no-load and load conditions	无裂纹、损伤、润滑良好 No cracks, damage, good lubrication 无异常振动、噪声和明显的发热 No abnormal vibration, noise and obvious heating
	滑动轴承 Slide bearing	检查轴承有无磨损；在空载和负载工况下是否烧损与发热 Check the bearings for wear and tear; burnout and heat generation under no-load and load conditions.	无明显磨损；不得有烧损或明显温升陡变 No obvious wear and tear; no burnout or significant temperature change
车轮组 Wheelset	轮缘 Rim	检查有无裂纹、缺蚀、变形、磨损 Check for cracks, chipping, deformation, wear and tear	无裂纹、缺蚀、明显变形、磨损 No cracks, chipping, obvious deformation, wear and tear
	轮毂及轮盘 Wheel hubs and wheel discs	检查有无裂纹、变形、磨损及损伤 Check for cracks, deformation, wear and damage	无裂纹、明显变形、磨损、损伤 No cracks, obvious deformation, wear, damage
	车轮踏面 Wheel tread	检查踏面有无磨损；主动车轮以及从动车轮直径误差；检查裂纹、变形、踏面表面剥落 Check the tread surface for wear and tear; driving wheel and driven wheel diameter error; check for cracks, deformation, tread surface spalling	无明显磨损；轮径误差值应符合相应标准；无裂纹与变形；无剥落 No obvious wear; wheel diameter error value in line with the standard; no cracks and deformation; no spalling
	轮毂内轴承 Inner hub bearing	检查滑动轴承的润滑情况；空载和负载工况时的异常振动、噪声、温升；	无异常 No abnormalities

		检查滚动轴承的润滑情况、振动、噪音、温升等 Check the lubrication of sliding bearings; abnormal vibration, noise and temperature rise under no-load and load conditions; check the lubrication, vibration, noise and temperature rise of rolling bearings.	正常 Normal
	车轮轮毂与端梁侧板之间的贴板 Pasteboard between the wheel hub and end beam side plate	检查有无摩擦、磨损；装配精度 Check for friction, wear and tear; assembly accuracy	无摩擦、磨损；安装良好 No friction, wear and tear; good installation

3.11 轨道的维护与保养
3.11 Track Maintenance and Servicing

3.11.1 经常检查起重机和小车的运行轨道的连接和固定螺栓，不得产生松动现象。
3.11.1 Frequently check the crane and trolley running track connection and fixing bolts for any loose

3.11.2 轨道上绝对禁止有油腻存在，必要时可在轨道上撒细砂等物以增加摩擦系数。
3.11.2 No grease on the track, if necessary, sprinkle fine sand and other things on the track to increase the friction coefficient.

3.11.3 轨道面在高低方向有波浪或在同一截面上两轨道高度差超差时应调整，使之达到规定要求。
3.11.3 Waves exist on the track surface or height difference in the same section exceeds the difference, adjustment should be carried out, so as to meet the specified requirements.

3.11.4 当轨道侧面磨损超过原轨道宽度的 15%，必须更换轨道。
3.11.4 When the track side wear exceeds 15% of the original track width, the track must be replaced.

轨道检查项目、内容及判定标准
Track Inspection Items, Content and Standards

检查项目 Inspection Items		检查内容 Inspection Contents	判定标准 Determination Standard
轨道 Track	钢轨 Steel track	检查有无裂纹、头部下陷、变形、侧面磨损 Check for cracks, head subsidence, deformation, side wear	无裂纹、明显下陷、变形及严重磨损 No cracks, obvious subsidence, deformation or serious wear and tear.
	钢轨紧固螺栓 Rail fastening bolts	检查连接螺栓有无松动及脱落 Check the connecting bolts for loosening and spalling	不得有松动与脱落 No loosening and spalling
	连接板及垫板 connecting plate and pad	检查螺栓有无松动、脱落；连接板和垫板有无移位、缺陷或脱落 Check the bolts for loosening and spalling;	无松动脱落；不得有移位、缺陷及脱落 No loosening and spalling; no

		connecting plates and pads for shifting, defects or spalling	shifting, defects and spalling.
	缓冲器及车挡 Buffer and car stop	检查有无损伤及错位；安装有无松动、脱落 Check for damage and misalignment; looseness and spalling in installation	无损伤、错位；无松动、脱落 No damage and misalignment; No looseness and spalling
	钢轨接头 Rail joint	检查钢轨接头有无错位及间隙变化 Check the rail joints for misalignment and clearance changes	不得有明显错位及间隙变化 no obvious misalignment and clearance changes
	钢轨焊接安装 Rail welding installation	检查焊缝有无裂纹与开裂 Check the weld for cracks	不得有裂纹、开裂 No crack is allowed.
	几何尺寸误差 Geometric size error	检查轨距、轨道中心线、轨顶高等偏差 Check the rail gauge, rail centerline, and rail elevation deviation	各偏差应不超过规范规定值 Deviation should not exceed the specification value.

3.12 限位开关的维护与保养
3.12 Maintenance and Servicing of Limit Switches

3.12.1 起重机装有安全开关、起升高度限位、各种极限位置、投料口中心等开关，保护起重机的安全运行。特别是自动控制状态的运行，起着决定性的作用，应定期检查各种开关的位置、灵活性等，以证明开关的可靠性。特别是制动器打开限位、起升限位器等开关由于部件磨损或更换后应保证原有的位置。
3.12.1 Cranes are equipped with safety switches, lifting height limit, various limit positions, feeding port center and other switches to protect the safe operation of the crane. The operation of the automatic control state, plays a decisive role. The position and flexibility of switches should be regularly checked to prove the reliability of the switch. In particular, switches such as the brake opening limit and the lift limiter should be secured in their original positions due to wear and tear of the parts or after replacement.

3.12.2 非检修维护人员禁止触摸各种限位开关，特别是有方向性的限位开关，如羊角开关等。
3.12.2 Non-overhaul-and-maintenance personnel are prohibited from touching limit switches, especially directional limit switches, such as horn switches.

3.12.3 操作人员不得依靠极限开关来代替自己的正常操作。
3.12.3 Operators shall not rely on limit switches as a substitute for their manual operation.

3.13 供电设备的维护与保养
3.13 Maintenance and Servicing of Power Supply Equipment

本项目的垃圾吊采用的移动供电方式是电缆滑车移动供电：电缆滑车方式经济（见图 1—5—1），架设容易，一般无需占用地面积。但无法中间点供电，软电缆较长，造成电力、信号能量损耗大，在环节差的地方，轨道容易卡死，造成滑车损坏（甚至电缆损坏）日常维护量较大。架设时滑车上电缆长度应该留有余度（电缆长度=1.15×S_1），当小车处于极限位置时，应使相邻两个滑车间的电缆保

持 120 度左右的夹角如下图。并使电缆长度较牵引钢丝绳略长些，使电缆除自身重量外避免承受额外拉力（S_1≤4m，钢丝绳=S_1-0.75 滑车）。垃圾吊的大、小车一般采用电缆滑车供电系统。
The waste crane of this project is powered by the cable pulley: cable pulley is economical (see Table 1-5-1), easy to set up, and in need of no land area. But cable pulley cannot supply power at intermediate points. Flexible cable is long, which leads to power loss and signal energy loss. When link is poor, the pulley is easy to be jammed, resulting in damage to the pulley (or even to the cable), which makes the daily maintenance more difficult. There should be some redundancy of the cable length on the pulley during erection (cable length = 1.15 × S_1). When the trolley is in the limit position, the cables between two adjacent pulley should be about 120 degrees as shown in the figure below. The cable length should be slightly longer than the traction wire, so to avoid additional tension of electric system other than its own weight (S_1 ≤ 4m, wire rope = S_1-0.75 pulley). The cranes and trolleys of the waste crane is powered by cable pulley power supply system.

3.14 各机构的日常维护
3.14 Routine Maintenance of Each Mechanism

各种检修周期的规定按起重机的工作级别及环境条件而定。垃圾抓斗起重机工作级别较高，必须建立完善的检修制度，保证设备的正常运行。
All kinds of maintenance cycle regulations are decided according to operation level and the environment. Waste grab cranes' working class are high, therefore, sound maintenance system must be established to ensure the normal operation of the equipment.

3.14.1 保持设备的环境卫生。观察减速箱是否漏油；
3.14.1 Keep the environment around the equipment clean. Check the gearbox for oil leakage.

3.14.2 用手探测电动机、减速箱的表面温度（重点在起升机构减速箱的输入、输出轴附近）。起升机构在额定速度运行的状态下，听减速箱、电动机的声音是否有异常；
3.14.2 Check the surface temperature of the motor and gearbox by hand (focusing on the input and output shaft of the lifting mechanism gearbox). When the lifting mechanism is running at the rated speed, see if the sound of the gearbox and motor is abnormal.

3.14.3 观察起升机构制动器推杆伸出长度是否满足要求（参照第四章“1”款）。
3.14.3 Check if the length of the lifting mechanism brake push rod meets the requirements or not (refer to Chapter IV, Clause 1).

3.14.4 观察三合一支撑点缓冲垫的完好情况，缓冲垫螺栓是否压紧。
3.14.4 Check the integrity of the buffer cushion of the three-in-one support point, and whether the buffer cushion bolts are tightened.

3.14.5 检查机械传动部件的轴、联轴器、制动轮毂等部件的完好情况，重点检查小车主动轮传动轴。
3.14.5 Check the integrity of shafts, couplings, brake hubs and other parts of

mechanical transmission components, with emphasis on the main wheel transmission shaft of the trolley.

3.14.6 观察大小车移动供电滑车、牵引滑车、牵引绳、电缆等的长度、间距是否均匀了、是否符合要求。
3.14.6 Check the length and spacing of the mobile power supply pulley, traction pulley, traction rope and cable of cranes and trolleys to see if they are even and meet the requirements.

3.14.7 钢丝绳与卷筒的连接固定情况，紧固连接部位的紧固件；
3.14.7 Wire rope and drum connection, fastening connection parts;

3.14.8 起升机构制动器推杆伸出长度每周进行一次调整。制动器动作是否对称，是否单边磨损严重；
3.14.8 The lifting mechanism brake push rod extension length is adjusted once a week. Whether brake action is symmetrical, one side of the wear and tear is serious;

3.14.9 抓斗油位、插头的连按和张紧释放装置；软管和螺纹接头与油缸的密封性；钢丝绳与抓斗的连接。
3.14.9 Grab oil level, the plug of the connecting press and tension release device; hose and threaded fittings and cylinder sealing; wire rope and grab connection.

3.14.10 测量起升机构制动器制动瓦块厚度是否满足要求；
3.14.10 Check the lifting mechanism brake tile thickness;

3.14.11 检查运行机构三合一制动器的制动性能，尤其是大车机构的制动器是否同步，制动力矩不宜过大，制动时不应有明显的金属撞击声（更不能出现车轮打滑）；
3.14.11 Check the braking performance of the three-in-one brake of the operating mechanism, especially whether the brake of the crane mechanism is synchronized, the braking torque should not be too large, and there should not be obvious metal impact sound (not to mention wheel slip) when braking.

3.14.12 每月注油一次（加油点、方法和油脂等见“润滑”章节）；
3.14.12 Oil should be injected once a month (see chapter "lubrication" for lubricanting points, methods and grease).

3.14.13 抓斗液压系统油压检测；
3.14.13 Check the lubricant pressure of the hydraulic system of the grab.

3.14.14 年大修—由机械工作（或专业）人员执行：年检修应列入工作计划，对一些部件应事先作好备品、备件工作。彻底解决日常维护中的一些问题，杜绝设备带病跨年度，造成重大危害。
3.14.14 Annual overhaul-by mechanical (or professional) personnel: annual repair should be included in the work plan, some parts should be made in advance. Thoroughly solve some of the problems in the daily maintenance. Faulty equipment should be fixed within the serving year to avoid more significant damage.

1）检查起重机大小车轮、轨道的几何形状和位置；
1) Check the geometry and position of the crane wheels and rails;

2）更换各机构减速箱齿轮油、油封。检查噪音和震动（重点是起升机构）；
2) Replace the gear lubricant and lubricant seals of the gearbox of each mechanism.

Check the noise and vibration (focusing on the lifting mechanism);.

3）检查各传动轴、联轴器、起升卷卷筒轴承座等机械传动部件的几何形状，是否有开焊、裂纹等问题。
3) Check the geometry of each transmission shaft, coupling, lifting drum, bearing seat and other mechanical transmission parts, whether there are open welding, cracks and other problems.

4）检查液压抓斗的抓齿等部位的磨损情况，着重检查抓斗与钢丝绳连接部位、连接用链环及抓齿。
4) Check the wear and tear of the hydraulic grab teeth and other parts of the grab, focusing on checking the grab and wire rope connection parts, connecting chain rings and grab teeth.

5）更换起升机构钢丝绳。
5) Replace the wire rope of the lifting mechanism.

6）桥架上拱度检测。
6) Check the camber of the bridge.

7）对油漆剥落处补漆、彻底清扫起重机上的油渍、灰尘等污物。
7) Repair the paint peeling place, thoroughly clean the oil stains, dust and other dirt on the crane.

3.15 电气设备的维护与保养
3.15 Maintenance and Servicing of Electrical Equipment

电气设备是垃圾吊的主要组成部分，其好坏直接影响设备的运行，电气设备经常有隐含的故障，只有加强日常维护才能使设备正常运行。
Electrical equipment is the main component of the waste crane, whose condition directly affects the operation of the equipment. Electrical equipment often have underlying faults, which could only operate normally under sound daily maintenance.

3.15.1 日检修—由起重机司机每天交接班时进行：清除电气设备外部的尘土、油污及其它附着物。用手探测接触器、控制器触头、电阻器等设备的发热情况。检查轴承有无漏油现象，主要设备的电线、电缆接头是否紧密。检查制动器、电动机的运行情况。在打开观察孔盖或外壳时，应防止灰尘等异物侵入设备内部（事后必须回复原密封闭状态）。将观察所得到的各种情况记录在“日志”上。
3.15.1 Daily overhaul-Performed by the crane driver at the daily shift: remove dust, lubricant, and other attachments from the exterior of the electrical equipment. Check the contactors, controller contacts, resistors and other equipment by hand. Check the bearings for oil leakage and the tightness of wire and cable joints of the main equipment. Check the operation of brakes and motors. When open the cover of the observation hole or the housing, dust and other exterior matter should be prevent from entering into the interior of the equipment (After the practice, the original closed state must restored). Record all the situations in the "journal".

3.15.2 旬检修或双周检修—由电气工作人员执行，司机也应参加：清除电气设备内部的尘土、油污及其它附着物。观察滑环箱的刷架、炭刷、滑环等的磨损情况。电动机、接触器、继电器及制动器等在运行时发出的声音是否正常，检查并修理

控制器与开关的触头，平整触头并涂上一层凡士林油。操纵人员配合机修人员调整制动器‘
3.15.2 Ten-day or two-week maintenance-performed by the electrical staff, the drivers should also be part of it: remove dust, oil stains, and other attachments from the interior of the electrical equipment. Check the wear and tear of brush holders, carbon brushes, slip rings, etc. in the slip ring box. Check the sound made by motors, contactors, relays and brakes, etc., during operation. Check and repair the contacts of controllers and switches, flatten the contacts and apply a coat of petroleum jelly. Operator works with mechanic to adjust brakes

3.15.3 年检修或大修—由电气工作人员执行:拆开各项电气设备进行清理，检修各项电气设备的支架。洗净电动机的滚动轴承并更换新润滑油脂，测量定子与转子之间的空隙，如发现不均匀时须更换滚动轴承。测量绝缘电阻，必要时应进行干燥。各种隐患、缺陷须在年修时全部修好，无法修好的部件应进行更换。年检修或大修应由各项设备的实际磨损与陈旧程度来决定其周期，不要过分拘泥时间。使用时应特别注意控制柜中的各主接触器的维护，及时消除故障，保持触头光滑、清洁、压力正常检查其机械联锁可靠性。在起重机上必须备有干式的灭火器。最常用的是四氯化碳灭器。不允许使用泡沫灭火器，干沙只能用来扑灭导线的着火，而不能用来扑灭电动机的着火。当发生火灾时，首先应设法切断电源，其它部位着火应用总低压断路器来切断电源。当保护盘面着火时，应切断馈电线上的上级开关。着过火的起重机要经过清擦，干燥与检修所有的电气设备及电气布线，修复合格后方能使用。
3.15.3 Annual repair or overhaul - performed by electrical staff: Disassemble the electrical equipment for cleaning and check each electrical equipment bracket. Clean the rolling bearings of the motor and replace them with new lubricating grease, measure the clearance between the stator and rotor, and replace the rolling bearings if unevenness is found. Measure insulation resistance and dry if necessary. All kinds of hidden dangers and defects should be repaired in the annual repair. Parts that cannot be repaired should be replaced. Annual repair or overhaul cycle should be decided by the actual wear and tear and years of the equipment. Do not be too rigid. Pay special attention to maintain main contactor of the control cabinet. Timely eliminate faults. Keep the contacts smooth, clean, and under normal pressure. Check the reliability of its mechanical interlock. Dry fire extinguishers must be equipped in the crane. The most commonly used is carbon tetrachloride extinguisher. Foam extinguishers are not allowed. Dry sand can only be used to extinguish the fire of the wires, not the fire of the motor. When a fire occurs, firstly cut off the power supply. When other parts get on fire, main low-voltage circuit breaker should be used to cut off the power. When the protection panel is on fire, the upper switch on the power feeder should be cut off. Cranes that have been on fire should be cleaned, dried and overhauled in regard to all electrical equipment and electrical wiring, and repaired before use.

3.15.4 只允许专业人员担任起重机电气的维护工作，并随时记录发生电气故障的日期、名称、发生故障的原因、检修方法及实际维修的问题等。
3.15.4 Only professional personnel are allowed to carry out crane electrical maintenance work. Record the date, name, cause, repair methods and the field maintenance problems of the failure.

3.15.5 如导电部分和裸线的外壳或保护罩不完整，则不允许使用起重机。
3.15.5 Cranes are not permitted to be used if the housings or protective covers for

conductive parts and bare wires are incomplete.

3.15.6 检修时必须采用电压在 36V 以下的便携式照明灯。
3.15.6 During the repair, portable lighting with a voltage under 36V must be there.

3.15.7 当必须带电工作时，一定要戴上橡胶手套，穿上橡胶靴并使用绝缘手柄的工具。另应有专人在一旦发生危险时立即切断电源。所有靠近导电部分的地方都必须用护栏围起来。
3.15.7 When work with electricity, always wear rubber gloves, rubber boots and use tools with insulated handles. Another person must be assigned to cut off the power supply as soon as any danger occurs. All areas near electrically conductive parts must be enclosed by guardrails.

3.15.8 电动机、电气设备和电气外壳上的所有金属部分可能发生导电的必须接地。
3.15.8 All metal parts on the motor, electrical equipment and electrical housing that may conduct electricity must be grounded.

起重机电气、控制系统检查项目、内容及判定标准
Crane Electrical, Control System Inspection Items, Content and Determination Standards

<table>
<tr><th colspan="3">检查项目
Inspection Items</th><th>检查内容
Inspection Contents</th><th>判定标准
Determination Standard</th></tr>
<tr><td rowspan="4">电动机
Motor</td><td colspan="2">绕组
Winding</td><td>检查绝缘电阻；有无发热
Check insulation resistance; and heating</td><td>绝缘电阻在规定范围；无异常发热
Insulation resistance in the specified range; no abnormal heating</td></tr>
<tr><td colspan="2">轴承
Bearing</td><td>检查润滑情况与异常响声
Check the lubrication and abnormal noise</td><td>润滑良好和无异常声响
Good lubrication and no abnormal noise</td></tr>
<tr><td colspan="2">滑环
Slip ring</td><td>检查有无变色、裂痕、接线头有无松动
Check whether there is discoloration, cracks, junction looseness</td><td>无明显变色、伤痕、破裂或松动
no obvious discoloration, scars, cracks or looseness</td></tr>
<tr><td colspan="2">电刷及导线
Brushes and wires</td><td>检查有无磨损和松动；压力；附着碳粉，转动电机轴有无火花
Check for wear and looseness; pressure; attached carbon powder, sparks when turning the motor shaft</td><td>无明显磨损、松动；压力适当；无附着碳粉；无火花
No obvious wear and tear, looseness; pressure is appropriate; no attached carbon powder; no sparks</td></tr>
<tr><td rowspan="2">集电装置
Power collector</td><td rowspan="2">滑线及滑车轨道
Slide line and pulley track</td><td>电源滑线、集电轨道
Power slide line, collector track</td><td>检查有无变形、磨损、损伤；张紧装置动作是否正常；滑线与滑块的接触情况；绝缘子支承有无松脱
Check whether there is deformation, wear and tear, damage; whether the action of the tensioning device is normal; the contact between the sliding line and the slider; whether the insulator support is loose.</td><td>无明显变形、磨损、损伤；张紧正常；接触良好；无松脱
No obvious deformation, wear, damage; normal tensioning; good contact; no looseness</td></tr>
<tr><td>壳、盖、罩子</td><td>检查有无损伤与变形；防触电装置是否正常</td><td>无损伤与明显变形；与滑线有足够间距</td></tr>
</table>

		Housing, cover, hood	Check for damage and deformation; whether the anti-shock device is normal	No damage and obvious deformation; having adequate spacing with the slide wire
		绝缘集电器 Insulated current collector	检查绝缘集电器的接线有无异常 Check whether the wiring of the insulated collector is abnormal	电缆心线、接头及外壳要可靠连接 The cable core, connector and housing should be connected reliably
		绝缘子 Insulator	检查有无脱落与松动、破裂与污垢 Check for spalling and loosening, cracking and dirt	无脱落松动、破裂与污垢 No spalling and loosening, cracking and dirt
	集电器 Current collector	机械部分 Mechanical Part	检查有无磨损与损伤；润滑是否良好 Check for wear and damage; good lubrication	无明显磨损、损伤；润滑良好 No obvious wear, damage; well-lubricated
		弹簧 Spring	检查有无变形、腐蚀及疲劳损伤 Check for deformation, corrosion and fatigue damage	无变形、无明显腐蚀及疲劳损伤 No deformation, no obvious corrosion and fatigue damage
		接线与绝缘 Wiring and insulation	检查接线有无断线，绝缘子是否破损、污秽 Check whether the wiring is broken, and whether the insulator is damaged or dirty	无断线或破损、污秽 No broken or damaged, dirty
		接头螺栓 Joint bolt	检查紧固部分有无松动脱落 Check the fastening part for spalling or loosening	无松动脱落 No loosening, spalling.
	供电电缆 Service cable	绝缘层 Insulating layer	检查有无损伤 Check for damage	无损伤 No damage
		连接处 Joint part	检查紧固部分有无松动与脱落 Check the fastening part for spalling or loosening	无松动脱落 No loosening, spalling.
		电缆及导向装置 Cable and guiding device	检查电缆拉伸部分有无弯曲、扭曲及损伤；电缆导向装置动作情况 Check whether the stretched part of the cable is bent, distorted and damaged; operation of cable guiding device	无弯曲、扭曲及损伤情况；动作平稳 No bending, distortion or damage; smooth operation
电气元件及控制系统 Electrical components and control systems	开关 Switch	开关、接触部分与保险器 Switches, contact parts and fuses	检查开关动作有无异常、外形有无破损；接触部分铰链和夹子的压力是否合适，保险器安装及容量是否合适 Check whether the switch operation is abnormal, shape is damaged; whether the pressure of the hinge and clamp of the contact part is appropriate, and whether the installation and capacity of the fuse are appropriate	动作正常、无破损；接触压力适当；安装正确，容量合适 Normal operation, no damage; appropriate contact pressure; appropriate installation and capacity
	接触器 Contactor	触头 Contact	检查触头接触压力及接触面破损 Check contact pressure and contact surface damage	接触面无间隙及脱开时彻底 Contact surface without clearance and completely removed

		弹簧 Spring	检查有无损坏、变形、腐蚀以及疲劳老化 Check for damage, deformation, corrosion and fatigue	无损坏变形，明显腐蚀和疲劳老化 No damage deformation, obvious corrosion and fatigue
		可动铁芯 Movable core	检查铁芯吸合面有无附着物；工作时有无异常声响，屏蔽线圈有无断线；检查限位块有无磨损及损伤；断路时有无间隙 Check whether the core suction surface is attached; whether there is no abnormal sound when working, and the shielding coil has no broken line; check the limit block for wear and damage; whether there is a clearance when the circuit is broken	无附着物；无异常声响或断线；无明显磨损及损伤；无间隙 No attachments; No abnormal sound or broken line; No obvious wear and damage; Zero clearance
		消弧线圈 Arc suppression coil	检查紧固部分有无松动 Check whether the fastening part is loose	无松动 No loose
		消弧栅 Arc suppression grid	检查是否在原位置；是否烧损 Check whether it is in the original position; burnt or not	应在原定位置；无明显烧损 Should be in the original location; no obvious burn damage
		紧固件 Fastening parts	检查有无松动 Check for loose	无松动 No loose
	继电器 Relay	弹簧 Spring	检查有无弯折、变形、腐蚀、疲劳损伤 Check for bending, deformation, corrosion, fatigue damage	无弯折、变形、明显腐蚀或疲劳损伤 No bending, deformation, obvious corrosion or fatigue damage
		机构及操作试验 Mechanism and operation test	用手操作，检查动作状态 Manually operate and check the action status	动作要正常 Normal action
电气元件及控制系统 Electrical components and control systems	电阻器 Resistor	动作方向显示板 Action direction display board	检查有无损伤及污染 Check for damage and stain	显示明显，无明显污损 The display is obvious, no obvious stain
		电线引入 Wire lead-in	检查电线引入管口有无异常 Check whether the wire inlet is abnormal	无损伤或明显变化 No damage or significant change
		端子 Terminal	检查紧固件有无松动 Check whether the fastener is loose.	无松动 No loose
		电阻片 Resistor	检查有无裂纹、损伤；各片间有无接触；有无松动；端子附近接线及绝缘是否过热烧损；绝缘体上是否积尘 Check for cracks and damage; whether there is contact between the tablets;	无裂纹、损伤；无接触；无松动；无烧坏；不得堆积粉尘 No crack, damage; no contact; no loosening; no

			whether it is loose; whether the wiring and insulation near the terminal are overheating and burning; whether there is dust accumulation on the insulator	burn out; no accumulation of dust
		绝缘子 Insulator	检查有无破裂、污损 Check for cracks and stains	无破裂与明显污损 No cracks and obvious stains
		连接紧固 Connection fastening	检查紧固件有无松动 Check whether the fastener is loose.	无松动 No loose
线路及通讯 Line and communication		机内明线 Internal open wire	检查保护层有无损伤；有无过紧、扭曲、线夹松动现象 Check the protective layer for damage; whether it is too tight, distorted or has loose clamp phenomenon	无损伤；不应过紧，扭曲、松动等 No damage; should not be too tight, distorted, loose, etc
		照明及信号灯 Lighting and signal lights	检查照明亮度是否合适；接头都分有无松动；紧固件是否松动；灯泡和防护装置有无破损 Check whether the lighting brightness is appropriate; whether the joint is loose or not; whether fasteners are loose; whether light bulbs and guards are broken or not	确保仪表和操作部分有充足的亮度；无松动；无破损 Ensure adequate brightness of the instrument and operating parts; no loosening; no damage
		通讯装置 Communication device	检查通话设施功能 Check communication facility functions	要求通话正常 Normal communication is required
		电路绝缘电阻 Circuit insulation resistance	测定配电电路各支路绝缘电阻有无异常 Whether the insulation resistance of each branch of the distribution circuit is normal	绝缘电阻值应在规定的范围之内 The insulation resistance value should be within the specified range
PLC		有无故障 Faulty or not	检查 PLC 程序是否正常运行 Check if the PLC program is running properly	运行灯常绿色，无闪烁 Running light always green, flicker free

第四章 调整
Chapter IV Adjustment

4.1 起升制动器的调整
4.1 Adjustment of Lifting Brake

4.1.1 推动器工作行程：松开锁紧螺母，转动拉杆，使推动器活塞杆：制动器断电闭合状态时留有 10~15%的总行程；制动器通电打开活塞杆推出到 85~90%时制动瓦块应完全打开，保证制动瓦块的退距；
4.1.1 Working stroke of the impeller: loosen the locknut and turn the pull rod to make the piston rod of the impeller: leave 10~15% of the total stroke when the brake is powered off and closed; When the piston rod is pushed out to 85~90%, the brake block should be fully opened to ensure the retreat of the brake block;

4.1.2 制动瓦块退距：制动瓦块与制动轮之间在制动器完全打开时的最大距离为 1.2mm，最小距离为 0.8mm。
4.1.2 Brake block retreat: The maximum distance between the brake block and the brake wheel when the brake is fully opened is 1.2mm and the minimum distance is 0.8mm.

4.1.3 制动瓦块均等：初次安装时，在制动器完全闭合状态，将均等连板拉起至水平位置，然后调整两侧螺钉并锁紧。此为自动均等装置，使用中无需调整。
4.1.3 Equal brake block: When the brake is fully closed for the first time, pull the equal connecting plate to the horizontal position, and then adjust the screws on both sides and lock it. This is an automatic equalizing device, without adjustment when using.

4.1.4 制动力矩：转动弹簧拉杆上端螺母可调节弹簧压缩长度，使调整螺钉的中心线对准所需制动力矩的刻度。
4.1.4 Braking torque: Turn the upper end nut of the spring rod to adjust the compression length of the spring, so that the center line of the adjusting screw is aligned with the scale of the desired braking torque.

4.2 运行“三合一”制动器的调整
4.2 Adjustment of the "three-in-one" Brake

4.2.1 制动力矩对于相应的型号其大小是固定不变的，出厂前已调好，并用火漆标记。客户不能自行调整。
4.2.1 The size of braking torque is fixed for the corresponding model, which has been adjusted before leaving the factory and marked with fire paint. Customers cannot make their own adjustments.

4.2.2 制动气隙的调整：拆下风罩，取下电机风扇；取出橡皮保护套；在断电条件下测量气隙；松开制动器后端的三个紧固用内六角螺栓；逆时针旋转调整螺母间隙变小，反之增大；调整后拧紧内六角螺栓；装回保护套、风扇、风罩。
4.2.2 Adjustment of brake air gap: remove the hood and motor fan; take out the rubber protective cover; measure the air gap under the condition of power failure; loosen the three fastening hex bolts at the rear end of the brake; turn counterclockwise to adjust

the nut gap becomes smaller, and vice versa increase; tighten the hex bolt after adjustment; reinstall the protective case, fan, hood.

4.2.3 抓斗开闭压力的调整
4.2.3 Adjustment of grab opening and closing pressure

按照抓斗的产品使用说明书进行调整。
Adjust according to the grab product use manual.

第五章 润滑
Chapter V Lubrication

起重机润滑的好坏，直接影响起重机各机构的正常运转，同时对延长机器零件的寿命和促进安全生产有密切关系。因此使用和维护人员必须经常检查各润滑点的润滑情况，按时补充和更换润滑油脂。

The quality of crane lubrication directly affects the normal operation of the crane institutions, and has a close relationship with the extension of the life of machine parts and the safe production. Therefore, the use and maintenance personnel must often check the lubrication of each lubrication point, and replenish and replace the lubricating oil on time.

5.1 润滑油润滑
5.1 Oil Lubrication

5.1.1 采用润滑油的润滑点：
5.1.1 Use Lubricating Points of Lubricating Oil:

1）液压抓斗油泵
1) Hydraulic grab oil pump

2）所有卷筒轴承座（箱）
2) All drum bearing seats (boxes)

3）起升减速器
3) Lifting reducer

4）运行“三合一”减速器
4) Run the "three-in-one" reducer

5）制动器用推动器
5) Actuator for brake

5.1.2 润滑油更换注意事项：
5.1.2 Precautions of lubricating oil replacement:

1）根据各零部件说明书的要求，在达到初始运转时间后，必须首次更换润滑油。润滑油应在停机后油温尚未降低时排放。
1) According to the requirements of the instructions of each part, the lubricating oil must be replaced for the first time after the initial running time is reached. The lubricating oil should be discharged when the oil temperature is not reduced after shutdown.

2）此后根据要求定期更换润滑油。当换油量较大时，可通过对润滑油进行化验的方法来确定最经济的换油间隔。
2) After that, replace the lubricating oil regularly as required. When the amount of oil change is large, the most economical oil change interval can be determined by testing the lubricating oil.

3）应换与原来同样等级的润滑油，不同等级或不同生产厂家的润滑油不能混合

使用。
3) It should be replaced with the same grade of lubricating oil as before, and the lubricating oil of different grades or different manufacturers can not be mixed.

4）换油时，箱体内部应采用与所用润滑油相同等级的油进行冲洗，高粘度油可先行预热再进行清洗。
4) When changing oil, the inside of the box should be washed with the same grade of oil, and the high-viscosity oil can be preheated before cleaning.

5）换油时，排油孔螺塞上的磁铁也要彻底清洗干净。
5) When changing oil, the magnet on the plug of the oil drain hole should also be thoroughly cleaned.

6）清洗必须绝对清洁，决不允许任何杂物进入机体内。
6) Absolutely clean, never allow any debris into the body.

7）清洗完毕，拧紧排油孔螺塞，再注入新油封好注油孔。
7) After cleaning, tighten the plug of the oil drain hole, and then inject new oil to seal the oil injection hole.

5.2 润滑脂润滑
5.2 Grease Lubrication

5.2.1 采用润滑脂的润滑点
5.2.1 Lubrication Points of Grease

1）液压抓斗节点和销轴
1) Hydraulic grab node and pin

2）钢丝绳
2) Steel wire rope

3）制动器各节点和销轴
3) Brake nodes and pins

4）大、小车供电导轨
4) Power supply rail of crane and trolly

5）卷筒联轴器
5) Drum coupling

6）轴承（包括电动机、车轮组、电缆卷筒等）
6) Bearing (including motor, wheelset, cable drum, etc.)

5.3 润滑脂添加注意事项
5.3 Precautions for Adding Grease

5.3.1 根据各零部件说明书的要求，定期对其添加润脂。
5.3.1 Add grease regularly according to the requirements of the instructions of each part.

5.3.2 润滑装置和润滑点必须清洁。润滑前应用浸有煤油的抹布清洗旧油。绝对禁止用金属刷或其它尖锐器具清洗污物；绝对禁止使用酸性或其它具有腐蚀性的润滑剂。
5.3.2 Lubrication devices and lubrication points must be clean. A kerosene-soaked rag should be used to clean the old oil before lubrication. It is absolutely prohibited to use metal brushes or other sharp utensils to clean dirt; the use of acidic or other corrosive lubricants is absolutely prohibited.

5.3.3 轴承应保持良好的润滑状态，注油量应为轴承体空腔的 2/3。
5.3.3 The bearing should be kept in good lubrication state, and the oil injection amount should be 2/3 of the bearing body.

5.3.4 不同的季节或环境应根据说明书的要求采用相应的润滑脂。
5.3.4 Different seasons or environments should use the corresponding grease according to the requirements of the manual.

典型零部件的润滑材料及其添加时间
Lubricating Materials of Typical Parts and Their Addition Time

序号 S/N	零部件名称 Name of Parts	润滑周期 Lubricating Cycle	润滑条件 Lubricating Condition	润滑材料 Lubricating Material
1	钢丝绳 Wire rope	一般 15～30 天一次，根据实际使用中的润滑情况而定 Generally once every 15 to 30 days, according to the actual use of lubrication	①把润滑脂加热到 50 一 100℃浸涂至饱和为止。 ① The grease is heated to 50 to 100℃ until saturated. ②不加热涂抹。 ② Apply without heat.	①丝绳麻心脂（SHO388 一 1992） ① Silk-rope hempwicking grease (SHO388 一 1992) ②石墨钙基润滑脂。 ② Graphite calcium base grease
2	减速器 Reducer	最初使用 100 小时后，清净内部，换上新的润滑油，以后每隔 2500 小时换油 After the first 100 hours of use, clean the interior, replace with new lubricating oil, and then change the oil every 2,500 hours	①油池飞溅润滑。 ① Oil pool splashing lubrication ②循环喷油润滑。 ② Circulating spraying lubrication	工业齿轮润滑油 N150、N320、N460、N680 按减速器说明书要求 Industrial gear oil N150, N320, N460, N680 according to the reducer instructions
3	开式齿轮 Open gear	半月一次，每季或半年清洗一次 Once every two months, once every quarter or half a year		明齿轮脂（HGI-26-73） Gear grease (HGI-26-73)
4	滚动轴承 Rolling bearing	3～6 个月一次 Once every three to six months	①工作温度在-20～120℃。 ① The operating temperature is -20 ~ 120℃. ②低于-20℃ ② Below -20℃	①通用锂基润滑脂 1、2、3 号（GB7324-1994） ① General purpose lithium lubricating grease 1, 2, 3 (GB7324-1994) ② 54 号低温润滑脂（SH0385-1992） ② No. 54 Low temperature grease (SH0385-1992)

第六章 故障及处理
Chapter VI Fault and Treatment

起重机械在使用过程中，机械零部件、电气控制和液压系统的元器件，不可避免地会发生有形磨损，并引发故障，导致同一故障的原因可能不是一一对应的关系。因此要对故障进行认真分析，准确地查找真正的故障原因，并采取相应的消除故障的方法来排除。从而恢复故障点的技术性能，以下为起重机常见故障及处理。

In the process of lifting appliance use, mechanical parts, electrical control and hydraulic system components, will inevitably occur physical wear, and cause failure, leading to the reason of the same failure possibly not closely associated with one another. Therefore, it is necessary to carefully analyze the fault, accurately find the real cause of the fault, and take the corresponding method to eliminate the fault. In order to restore the technical performance of the fault point, the following is the common fault and treatment of the crane.

6.1 自动报警故障说明
6.1 Automatic Alarm Fault Description

序号 S/N	故障名称 Fault Name （PLC 的采样点）工作形式 (PLC sampling point) working form	发生故障原因 Cause of Fault	解除故障的方法 Troubleshooting Methods
1	小车运行启动失败 Trolley failed to start	1．接触器线圈断路，触点卡住 1. The contactor coil is disconnected and the contact is stuck 2．接触器线圈控制回路断路 2. Control circuit of contactor coil disconnected 3．信号传输回路断路 3. The signal transmission circuit is disconnected	1．检修或更换接触器 1. Repair or replace the contactor 2．检修接触器的控制回路 2. Check the control circuit of the contactor 3．检修信号传输回路 3. Check the signal transmission circuit
2	大车运行启动失败 Crane failed to start	1．接触器线圈断路，触点卡住 1. The contactor coil is disconnected and the contact is stuck 2．接触器线圈控制回路断路 2. Control circuit of contactor coil disconnected 3．信号传输回路断路 3. The signal transmission circuit is disconnected	1．检修或更换接触器 1. Repair or replace the contactor 2．检修接触器的控制回路 2. Check the control circuit of the contactor 3．检修信号传输回路 3. Check the signal transmission circuit
3	起升机构启动失败 The lifting mechanism failed to start	1．制动器电动机损坏无法打开 1. The brake motor is damaged and cannot be opened 2．制动器液压油泄漏无力找开 2. Brake hydraulic oil leaks, causing weakness 3．制动器闸瓦过渡磨损或行程开关机构失调致使开关不合 3. Brake shoe excessive wear or stroke limit switch mechanism	1．更换制动器 1. Replace brake 2．补充液压油 2. Refill hydraulic oil 3．更换闸瓦、调整行程开关机构 3. Replace the brake shoe and adjust the stroke switch mechanism

6		misalignment resulting in switch not closing 4．行程开关杨构、触点损坏 4. Stroke switch contact is damaged 5．电动机供电回路断路制动器不动作 5. Motor supply circuit is disconnected, brake does not operate 6．外信号回路断路不动作继电器线圈、 6. The external signal circuit is disconnected and the relay coil does not operate 7．继电器触点损坏或其控制电路断路 7. The relay contact is damaged or its control circuit is disconnected 8．制动器接触器、线圈、触点损坏或其控制电路断路 8. Brake contactors, coils, contacts are damaged or their control circuits are disconnected	4．更换机构或行程开关 4. Replace the mechanism or the stroke switch 5. 检修制动器电动机供电回路 5. Check the brake motor supply circuit 6．检修外信号回路 6. Overhaul the external signal circuit 7．更换继电器 7. Replace the relay 8. 检修或更换制动器接触器 8. Repair or replace the brake contactor
4	小车变频器故障 Fault of trolley converter	1．变频器故障根据变频器使用手册查询故障分析故障发生的原因。 1. Faulty of the frequency converter. Query the fault according to the converter use manual and analyze the cause of the fault. 2．造成变频器故障的其它原因如电机损坏、制动器不能打开、动力电缆断路、制动器控制电路断路等。 2. Other causes of converter failure such as motor damage, brake failure to be opened, power cable disconnected, brake control circuit disconnected, etc.	1. 由专业人员调整检修变频器。 1. The converter is adjusted and repaired by professional personnel. 2．检修更换电动机、制动器及其动力、控制电路。 2. Overhaul and replace the motor, brake and its power and control circuits.
5	大车变频器故障 Fault of crane converter	1．变频器故障根据变频器使用手册查询故障分析故障发生的原因。 1. Faulty of the frequency converter. Query the fault according to the converter use manual and analyze the cause of the fault. 2．造成变频器故障的其它原因如电动机损坏、制动器不能打开、动力电缆断路、制动器控制电路断路等。 2. Other causes of converter failure such as motor damage, the brake cannot be opened, power cable disconnection, brake control circuit disconnection.	1. 由专业人员调整检修变频器。 1. The converter is adjusted and repaired by professional personnel. 2．检修更换电动机、制动器及其动力、控制电路。 2. Overhaul and replace the motor, brake and its power and control circuits.
6	起升机构变频器故障	1．变频器故障根据变频器使用手	1. 由专业人员调整检修变频

	Fault of the converter of the lifting mechanism	册查询故障分析故障发生的原因。 1. Faulty of the frequency converter. Query the fault according to the converter use manual and analyze the cause of the fault. 2．造成变频器故障的其它原因如电动机损坏、制动器不能打开、动力电缆断路、制动器控制电路断路等。 2. Other causes of converter failure such as motor damage, the brake cannot be opened, power cable disconnection, brake control circuit disconnection.	器。 1. The converter is adjusted and repaired by professional personnel. 2．检修更换电动机、制动器及其动力、控制电路。 2. Overhaul and replace the motor, brake and its power and control circuits.
7	超 载 110%（=31—K632）载荷110%K632 闭合 Overload 110% (=31-K632) Load 110% K632 closed	1. 载荷确实达到 110%起重机停止工作 1. Load does reach 110% crane stops working 2．称重系统故障 2. Weighing system failures 3．信号线路故障 3. Signal line failure	1. 降低抓斗高度打开抓斗重新抓取 1. Lower the grab height to open the grab to re-grab 2. 请专业人员调整称重系统 2. Have the weighing system professionally adjusted 3．检修信号线路 3. Overhaul of signal lines
9	门开关未关或故障 Door switches left open or malfunctioning	1．未关门 1. Keep the door open 2．未供电 2. Unpowered 3．限位开关损坏 3. Damaged limit switches	1．关门 1. Closed door 2．检查线路 2. Check the circuit 3．更换限位开关 3. Replacement of limit switches
10	钥匙开关故障 Key Switch Fault	1．钥匙开关未合 1. Key switch open 2．触电损坏或线路故障 2. Electrocution damage or wiring failure 3．相应继电器损坏 3. Damage to the corresponding relay	1．转动钥匙，闭合 1. Turn key to close 2．更换触点或查线路 2. Replacement of contacts or wiring check 3．更换继电器 3. Replacement of relays
11	抓斗配电故障 Grab power distribution failure	1．未合闸 1. Switch off 2．保护跳掉 2. Leakage protector tripped 3．断路器损坏 3. Damage to circuit breakers	1．合闸 1. Switch on 2. 重合或监控电流并调整电流值 2. Reclosing or monitoring the current and adjusting the current value 3．更换断路器 3. Replacement of circuit breakers

6.2 常见故障及处理
6.2 Common Faults and Treatment

6.2.1 操纵线路故障
6.2.1 Control line failure

故障情况 Fault	发生故障的原因 Causes of Fault	消除故障的方法 Methods of Eliminating Faults
合上总低压断路器关合钥匙开关按下起动接钮 Close the main low voltage circuit breaker, close the key switch, and press the start button. 之后控制回路断路器 After that, control circuit breakers	1．控制回路有接地或短路故障 1. Grounding or short-circuit failure in the control circuit 2．主接触器线圈接地或匝间短路 2. Main contactor coil grounding or turn-to-turn short circuit	1．排除接地或短路故障 1. Remove grounding or short-circuit failures 2．检修或更换主接触器的线圈 2. Repair or replace the coil of the main contactor
按下起动接钮主接触 Press the master contact of starter button 器不吸合 The device does not engage	1．接触器线圈断路 1. Contactor coil disconnection 2．门开关连锁触头接触不良 2. Poor door switch interlock contacts 3．线路断路或接线端松脱 3. Disconnected line or loose terminal 4．接触器机件卡住或弹簧力过大 4. The contactor mechanism is stuck or the spring force is too large.	1．检修或更换线圈 1. Overhaul or replace the coil 2．检修接触不良的触头 2. Overhaul of poor contacts 3.接好断线或宁紧松动的接线端 3. Connect broken or loose terminals. 4.排除机件故障后调整弹簧张力 4. Adjust the spring tension after troubleshooting the machine.
当终端开关动作时相应的电动机不断电 The corresponding motor is continuously charged when the terminal switch is operated	1．终端开关的电路中发生短路故障 1. There is a short circuit failure in the circuit of the terminal switch 2．控制线路的导线次序错乱 2. The wires of the control line are out of sequence	1．检查检修排除短路导线 1. Check and repair to remove short-circuit wires 2．检查接线 2. Check the wiring
电源切断后接触器不掉下 The contactor will not fall off after the power is cut off	1．操作电路中接地或短路 1. Grounding or short-circuiting in operating circuits 2．接触器触头焊住 2. Contactor contacts welded	1．查找接地或短路点进行处理 1. Finding grounding or short-circuit points for treatment 2．检修或更换接触器 2. Repair or replace the contactor

6.2.2 控制器
6.2.2 Controller

故障情况 Fault	发生故障的原因 Causes of Fault	消除故障的方法 Methods of Eliminating Faults
合上总低压断路器关合钥匙开关按下起动接钮之后控制回路断路器 Close the main low voltage circuit breaker, close the key switch, and press the starter switch. After that, control circuit breakers	1．控制回路有接地或短路故障 1. Grounding or short-circuit failure in the control circuit 2．主接触器线圈接地或匝间短路 2. Main contactor coil grounding or turn-to-turn short circuit	1．排除接地或短路故障 1. Remove grounding or short-circuit failures 2．检修或更换主接触器的线圈 2. Repair or replace the coil of the main contactor
按下起动接钮主接触器不吸合 Press the master contact	1．接触器线圈断路 1. Contactor coil disconnection 2．门开关连锁触头接触不良	1．检修或更换线圈 1. Overhaul or replace the coil 2．检修接触不良的触头

of starter button, the device does not engage	2. Poor door switch interlock contacts 3．线路断路或接线端松脱 3. Disconnected line or loose terminal 4．接触器机件卡住或弹簧力过大 4. The contactor mechanism is stuck or the spring force is too large.	2. Overhaul of poor contacts 3.接好断线或宁紧松动的接线端 3. Connect broken or loose terminals. 4.排除机件故障后调整弹簧张力 4. Adjust the spring tension after troubleshooting the machine.
当终端开关动作时相应的电动机不断电 The corresponding motor is continuously charged when the terminal switch is operated	1．终端开关的电路中发生短路故障 1. There is a short circuit failure in the circuit of the terminal switch 2．控制线路的导线次序错乱 2. The wires of the control line are out of sequence	1．检查检修排除短路导线 1. Check and repair to remove short-circuit wires 2．检查接线 2. Check the wiring
电源切断后接触器不掉下 The contactor will not fall off after the power is cut off	1．操作电路中接地或短路 1. Grounding or short-circuiting in operating circuits 2．接触器触头焊住 2. Contactor contacts welded	1．查找接地或短路点处理 1. Finding grounding or short-circuit points to deal with 2．检修或更换接触器 2. Repair or replace the contactor
控制器在工作中产生卡住现象 Controller is jammed during operation	1．触头焊住 1. Contacts welded 2．定位机构发生故障 2. Fault of the positioning mechanism	1．修复或更换触头 1. Repair or replacement of contacts 2．检查并修复定位机构 2. Inspection and repair of positioning mechanisms
触头严重烧损 Severe burns on contacts	1．触头接触不良 1. Poor contact 2．控制器过载 2. Controller overload	1.调整触头的压力清除触头表面的赃物 1. Adjust the contact pressure to remove the dirt on the contact surface.

6.2.3 交流电动机
6.2.3 AC motors

故障情况 Fault	发生故障的原因 Causes of Fault	消除故障的方法 Methods of Eliminating Faults
整个电动机均匀地过热 Uniform overheating of the entire motor	1．工作级别超出设计值 1. Working level exceeds design value 2．电动机的工作电压过高或过低 2. The working voltage of the motor is too high or too low. 3．制动器打开时间隙过小或闸瓦不对称 3. The brake opens with too little clearance or asymmetrical brake shoe.	1．降低工作级别或更换电动机 1. Reduce the working level or replace the motor 2.调整供电电源或在电动机起动后检查供电线路中的压降，根据情况更换导线压紧接线端子 2. Adjust the power supply or check the voltage drop in the power supply line after starting the motor, and replace the wires and press the terminals accordingly. 3．调整检修制动器 3. Adjustment of service brakes
定子铁芯局部过热 Localized overheating of stator core	铁芯的矽钢片间发生局部短路 Localized short circuit between silicon steel sheets of the iron core	消除毛刺或其它引起短路的地方然 Eliminate burrs or other areas that

		cause short circuits, 后涂以绝缘漆 then apply insulating varnish
电动机不能起动 Motor does not start	1．定子绕组或电源电路有一相开路 1. One phase of the stator winding or power supply circuit is open-circuited 2．定子绕组接线错误或相间短路、接地 2. Stator winding wiring error or phase short circuit, grounding 3．鼠笼式转子电动机的转子有断条 3. Squirrel-cage rotor motors with broken rotor bars	1．检查并排除断相的原因 1. Check and eliminate the cause of phase failure 2．纠正接线，检查短路、接地的部位加以修复 2. Correct wiring, check for short circuits and grounded parts and repair them. 3．修复断条或更换电动机 3. Repair broken bars or replace motor
电动机在工作时振动 Motor vibration during operation	1．电动机轴和减速机轴不同心 1. Motor shaft and reducer shaft are not concentric 2．轴承磨损 2. Bearing wear 3．转子变形、动平衡失调 3. Rotor deformation, dynamic balance disorders	1．纠正同心度 1. Correcting concentricity 2．检查、检修或更换轴承 2. Inspect, overhaul or replace bearings 3．调整变形，重新进行动平衡检测或更换电动机 3. Adjust the deformation, re-perform the dynamic balance test or replace the motor
电动机在工作中声音不正常 Motor sounds abnormal during operation	1．定子相位错移 1. Stator phase misalignment 2．定子铁芯未压紧 2. Stator core not compressed 3．轴承磨损 3. Bearing wear 4．槽楔膨胀 4. Slot wedge expansion	1．检查联接线系统 1. Inspection of the connecting line system 2．检查定子，重新压装定子铁芯 2. Check the stator and re-press the stator core. 3．检查更换轴承 Inspect or replace bearings

6.2.4 交流液压推动器
6.2.4 AC hydraulic propeller

故障情况 Fault	发生故障的原因 Causes of Fault	消除故障的方法 Methods of Eliminating Faults
通电后不动作或推力不足、行程小 No action or insufficient thrust or small stroke after being energized	1．电动机缺相或绕组短路，接线端子接触不良 1. Motor lack of phase or winding short circuit, poor contact with the terminals 2．供电电压低于额定值 2. Supply voltage below the rated value 3．严重漏油或油量不足 3. Serious oil leakage or insufficient oil	1. 消除缺相、更换电动机或压紧接线端子 1. Eliminate phase loss, replace motor, or compress terminals 2. 检查电源电压及造成压降的原因 2. Check the power supply voltage and the cause of voltage drop 3．消除漏油、补充油液 3. Elimination of oil leaks, replenishment of oil
电动机声音不正常 Abnormal motor sound	轴承磨损，定子、转子气隙不均 Worn bearings, uneven stator and rotor air gap	更换轴承，检修定子、转子的气隙配合 Replacement of bearings, overhaul

		of stator and rotor air gap fits
电动机温升过高 Excessive temperature rise in motors	定子、转子相擦，供电电压过 Stator and rotor rubbing together, supply voltage too low or high 低或过高，电动机受潮 Moisture in motors	消除定子、转子的摩擦现象，检查调整电源电压，电动机进行干燥处理 Eliminate the friction phenomenon of stator and rotor, check and adjust the power supply voltage, and dry the motor.

6.2.5 交流接触器及继电器

6.2.5 AC contactors and relays

故障情况 Fault	发生故障的原因 Causes of Fault	消除故障的方法 Methods of Eliminating Faults
线圈温度过高 High coil temperature	1．由于种种因素动、静铁芯接触不上 1. Due to a variety of factors dynamic and static iron core contact is not available 2．线圈匝间短路或电压等级不对 2. Coil turn-to-turn short circuit or wrong voltage level	1.消除引起铁芯接触不上的原因 1. Eliminate causes of poor iron core contact （机件卡住、铁芯弯曲、赃物） (machine parts jammed, iron core bent, dirt) 2.检修或更换线圈（注意电压等级） 2. Overhaul or replace the coil (note the voltage level)
产生较大的噪音 Produces loud noise	1．线圈过载（电流过大） 1. Coil overload (excessive current) 2．动、静铁芯工作表面脏污 2. Dirty working surface of dynamic and static iron core 3．磁路弯曲 3. Magnetic circuit bending 4．短路环开路 4. Short-circuit loop open	1．减小动触头弹簧的压力 1. Reduce the pressure on the spring of the dynamic contact. 2．消除脏污 2. Elimination of dirt 3．调整动静触头的位置 3. Adjustment of the position of the dynamic and static contacts 4．修复短路环或更换铁芯 4. Repair the short-circuit loop or replace the iron core
动作迟缓 Delayed movement	1．动铁芯离静铁芯过远 1. The dynamic iron core is too far away from the static iron core. 2．器械底板的上部教下部凸出 2. The upper part of the machine base plate protrudes from the lower part	1．缩短两部分间的距离 1. Reduce the distance between the two parts 2．垂直装置器械 2. Vertical installation machines
断电时铁芯不掉下 The iron core does not fall down when the power is cut off	1．触头的压力不够 1. Insufficient contact pressure 2．动铁芯或机械部分被卡住 2. The dynamic iron core or mechanical parts are stuck. 3．触头焊接在一起 3. Contacts welded together	1．增加触头压力 1. Increase contact pressure 2．消除障碍物 2. Elimination of obstacles 3.更换触头并排除触头熔焊的原因 3. Replacement of contacts and elimination of causes of contact fusion welding
触头过热或烧焦 Overheated or burnt contacts	1．触头压力过小 1. Low contact pressure 2．触头脏污 2. Dirty contacts	1．调整弹簧压力 1. Adjustment of spring pressure 2．清除脏污或更换触头滑块 2. Remove dirt or replace contact

		sliders

6.2.6 变频器故障

6.2.6 Frequency converter failure

故障情况 Fault	发生故障的原因 Causes of Fault	消除故障的方法 Methods of Eliminating Faults
变频器失控，指令取消后变频器继续输出 Frequency converter is out of control, the frequency converter continues to output after the command is canceled	1．变频器控制信号错乱 1. Frequency converter control signal error 2．变频器控制程序错乱 2. Frequency converter control program error	1．紧急停止，关闭电源检查调整控制信号电路 1. Emergency stop, power off to check and adjust control signal line 2．紧急停止，关闭电源请专业人员检修变频器 2. Emergency stop, power off and ask a professional to repair the converter
变频器直流母排过压故障 Frequency converter DC bus over-voltage failure	1．制动电阻器断路，电动机回馈能量无处泄放 1. The braking resistor is disconnected, and there is no place for the motor's return energy to be discharged. 2．电动机过载或工作过于频繁使电阻器过热电阻率升高 2. Motor overload or work too frequently so that the resistor overheating and resistivity increases	1．更换制动电阻器 1. Replace the brake resistor 2．减少载荷或降低工作频率，降温 2. Reduce load or operating frequency and reduce temperature
变压器过热、过流、过压故障 Transformer overheating, overcurrent, overvoltage failure	1．环境温度超标（40 度） 1. Excessive environmental temperature (40 degrees) 2．过载使用 2. Overloading 3．电网电压波动超标 3. Excessive grid voltage fluctuations 4．变频器故障 4. Frequency converter failure	1．降低环境温度 1. Lowering the environmental temperature 2．减轻载荷 2. Load reduction 3．检查电网 3. Inspection of the grid 4. 由专业人员调整或更换变频器 4. Adjustment or replacement of the converter by professional staff
变频器制动单元烧毁 Converter brake unit burned out	1．制动电阻器及回路短路 1. Braking resistor and circuit short 2．频繁重载下放 2. Frequent reloading and decentralization	a）更换制动单元及制动电阻器 a) Replacement of brake units and brake resistors b)更换制动单元并降低工作频率 b) Replacement of the brake unit and reduction of the operating frequency

6.2.7 交流接触器及继电器

6.2.7 AC contactors and relays

故障情况 Fault	发生故障的原因 Causes of Fault	消除故障的方法 Methods of Eliminating Faults
线圈温度过高 High coil temperature	1．由于种种因素动、静铁芯接触不上 1. Due to a variety of factors	1. 消除引起铁芯接触不上的原因（机件卡住、铁芯弯曲、赃物） 1. Elimination of causes of poor

	dynamic and static iron core contact is not available 2．线圈匝间短路或电压等级不对 2. Coil turn-to-turn short circuit or wrong voltage level	iron core contact (jammed parts, bent iron core, dirt) 2.换线圈（注意电压等级） 2. Change the coil (note the voltage level)

6.3 金属结构部分故障
6.3 Partial Fault of Metal Structures

（1）金属结构部分故障（表 6-3-1）
(l) Partial failure of metal structures (Table 6-3-1)

表 6-3-1 金属结构部分故障
Table 6-3-1 Metal Structure Partial Fault

故障名称 Fault Name	故障原因 Fault Cause	排除方法 Troubleshooting Method
主梁腹板或盖板发生疲劳裂纹 Fatigue cracking of main beam webs or covers	长期超载使用 Long-term overloaded use	裂纹不大于 0.lmm 的，可用砂轮将其磨平，对于较大的裂纹，可在裂纹两端钻大于Φ8-的小孔，然后沿裂纹两侧开 60°的坡口，进行补焊。重要受力构件部位应用加强板补焊，以保证其强度。 If the crack is not larger than 0.1mm, it can be smoothed by grinding wheel. For larger cracks, small holes larger than Φ8- can be drilled at both ends of the cracks, and then a 60° groove is opened along both sides of the cracks to carry out the filler welding. Stiffening plates should be used to reinforce the parts of important stress components in order to ensure their strength.
主梁腹板有波浪形变形 Wavy deformation of main beam web	超负荷使用，使腹板局部失稳 Overloading and localized instability of the web plate	采取火焰矫正，消除变形，锤击消去内应力，严禁超负荷使用 Adopt flame rectification to eliminate deformation, hammering to eliminate internal stress, overloading is strictly prohibited.
主梁旁弯变形 Main beam side bending deformation	工作应力叠加所致，运输和存放不当 Improper transportation and storage due to superimposed operational stresses	用火焰矫正法，在主梁的凸起侧加热，并适当配用顶具和拉具 Flame rectification method, heating on the raised side of the main beam, with proper use of jacking and pulling tools
主梁下沉变形 Main beam subsidence deformation	主梁结构应力腹板波浪形变形超载使用热效应的影响存放、运输不当及其他 Structural stress of the main beam wavy deformation of the web, the influence of thermal effects of overloaded use. Improper storage, transportation and other	采用预应力法矫正，采用火焰矫正后，并沿主梁下盖板用槽钢加固 Rectification by electrothermal prestressing method, after using flame rectification and reinforced with channels along the lower cover plate of the main beam

（2）各机构的故障及排除方法（表 6-3-2、表 6-3-3、表 6-3-4）
(2) Faults and troubleshooting methods for each mechanism (Table 6-3-2, Table 6-3-3, Table 6-3-4)

表 6-3-2 桥式起重机各机构故障（起升机构各零部件故障）及排除方法

Table 6-3-2 Fault of Each Mechanism of the Bridge Crane (failure of each part of the lifting mechanism) and Troubleshooting Methods

零件名称 Part Name	故障及损坏情况 Faults and Damages	原因与后果 Causes and Consequences	排除方法 Troubleshooting Method
减速器 Reducer	过热 Overheating	1．负载过大 1. Overloading 2．润滑油不足或过热 2. Insufficient or overheated lubricating oil	检查实际负载，调整到规定值上或更换交大功率的减速器 Check the actual load and adjust it to the specified value or replace the reducer for higher power transmission
	异常的稳定运转噪声 Abnormal stable operating noise	1.转动、研磨噪声、轴承损坏 1. Rotation, grinding noise, bearing damage 2.敲击噪声，啮合不规则 2. Knocking noise, irregular meshing	1 检查润滑油 1. Check the lubricating oil 2、与客户服务机构联系 2. Contact the customer service agency
	异常的不稳定运转噪声 Abnormal unstable operating noise	油污染或油量不足 Oil fouling or insufficient oil volume	换油或加油至规定值 Change the oil or refuel to the specified value
	通气器漏油 Oil leakage from the ventilator	1、油量太多 1. Too much oil 2、通气器安装不正确 2. Incorrect installation of the ventilator	1、修正油位 1. Correct the oil level 2、正确安装通气器 2. Correct installation of the ventilator
	漏油 Oil leakage	密封损坏 Sealing part is damaged	与用户服务机构联系 Contact the user service agency
电动机 Motor	转动时输出轴不转 The output shaft does not rotate when turning	减速器键联接破坏 The reducer key coupling is broken	送去检修 Sent for maintenance
钢丝绳 Wire rope	断丝、断股、打结、磨损 Broken wires, broken strands, knots, wear and tear	导致突然断绳 Result in a sudden rope breakage	断股、打结停止使用；断丝，按标准更换；磨损，按标准更换 Broken strands and knots need to be stopped; broken wire, replaced as standard; Wear and tear, replaced as standard
轴 Shaft	1、裂纹 1. Crack 2、轴弯曲超过0.5mm/m	1、材质差，热处理不当，导致损坏轴 1. Poor material and improper heat treatment lead to shaft damage	1、更换 1. Replace 2、更换或校正 2. Replace or correct 3、起升机构应作更换，运行机构等可修

	2. Axis bending more than 0.5mm/m 3、键槽损坏 3. Key slot is damaged	2、导致轴颈磨损，影响传动，产生振动 2. Cervical wear and tear, affecting transmission, and producing vibration 3、不能传递扭矩 3. Torque cannot be transmitted	复使用 3. The lifting mechanism should be replaced, and the operating mechanism can be repaired and used
车轮 Wheel	1、踏面和轮辐轮盘疲劳裂纹 1. Fatigue cracks on tread spoke rim 2、主动车轮踏面磨损不均匀 2. The driving wheel tread is worn unevenly 3、踏面磨损达轮圈厚度的15% 3. Tread wears up to 15% of rim thickness 4、轮缘磨损达原厚度的 50% 4. Flange wears up to 50% of original thickness	1、车轮损坏 1. Wheel damage 2、导致车轮啃轨，车体倾斜和运行时产生振动 2. Cause wheels to gnaw on the rails, tilt the car body and vibrate during operation 3、车轮损坏 3. Wheel damage 4、由车体倾斜、车轮啃轨所致，容易脱轨 4. It is caused by the tilting of the car body and the gnawing of the wheels, which is easy to derail	1、更换 1. Replacement 2、成对更换 2. Pair replacement 3、更换 3. Replacement 4、更换 4. Replacement
滚动轴承 Rolling bearing	温度过高 Overheating 1.异常声响（断续的哑音） 1. Abnormal sounds (intermittent muffled sounds) 2.金属研磨声响 2. Metal grinding sound 3.锉齿声或	1.润滑油污垢完全缺油或油过多 1. Lubricating oil fouling is completely devoid of oil or too much oil 2.轴承污脏 2. Dirty bearings 3.缺油 3. Oil shortage 4.轴承保持架、滚动体损坏 4. Damage to bearing cages and rolling elements	1.清除污垢，更换轴承，检查润滑油数量 1. Remove dirt, replace bearings, check the amount of lubricating oil 2.清污 2. Remove dirt 3.加油 3. Add lubricating oil 4.更换轴承 4. Change the bearing

	冲击声 3. Filing or impacting sound		
滑动轴承 Slide bearing	过度发热 Overheating	1、轴承偏斜或压得过紧 1. The bearing is misaligned or overly compressed 2、间隙不当 3、润滑剂不足 2. Improper clearance 3. Insufficient lubricant 4、润滑剂质量不合格 4. Failed lubricant quality	1、消除偏斜，合理紧固 1. Eliminate deflection and tighten properly 2、调整间隙 2. Adjust the clearance 3、加润滑油 3. Add lubricating oil 4、换合格的油剂 4. Replace qualified lubricating oil

表 6-3-3 桥式起重机故障及排除（起升机构部件部分）

Table 6-3-3: Faults and Troubleshooting for Bridge Cranes (Lifting Mechanism Parts)

故障 Fault	主要原因 Main Causes	处理方法 Treatment Method
起动后，电机不转，不能起吊重物。 After starting, the motor does not rotate and cannot lift heavy objects.	过度超载 Overloading	不允许超载使用 Overloading is not allowed
	电源进线相序不对，主接触器不能吸合 The power incoming phase sequence is incorrect, and the main contactor cannot close	任意调换两根电源进线 Exchange the two power supply inlet lines arbitrarily
	电压比额定电压低 15％以上 The voltage is more than 15% below the rated voltage	等电压恢复正常 Wait for the voltage to normalize
	电器有故障，导线断开或接触不良 The appliance is malfunctioning, the wire is disconnected or the contact is poor	检修电器与线路 Overhaul appliances and wiring
	制动轮与后端盖锈蚀咬死，制动轮脱不开 The brake wheels and the rear end cover are rusted and bitten, and the brake wheel cannot be separated	卸下制动轮，清洗锈蚀表面 Remove the brake wheels and clean the rusted surface
漏油 Oil leakage	箱体盖之间，密封圈装配不良或失效损坏 Between the box covers, the sealing ring is poorly assembled or damaged by failure	拆下检修或更换密封圈 Remove for service or replace the seal
	联接螺钉未拧紧 The coupling screws are not tightened	拧紧螺钉 Tighten the bolts
电动机温升过高 Excessive temperature rise in motors	超载使用 Overloaded use	不允许 Not allowed
	作业过于频繁 Operation is too frequent	按 FC40％工作制 Based on FC 40% working system
	制动器间隙太小，运转时制动环未完全脱开，相当于附加载荷 The brake clearance is too small, the	重新调整间隙 Readjust the clearance

	brake ring is not completely disengaged during operation, which is equivalent to additional load	
起动时电机发出嗡嗡声 The motor makes a buzzing sound when starting up	电源及电机少相 Phase loss of the power supply and motor	检修或更换接触器 Overhaul or replace contactors
	交流接触器接触不良 Poor contact of AC contactors	
电动机转动时输出轴不转 The output shaft does not rotate when the motor rotates	减速器键联接破坏 The reducer key coupling is broken	送去检修 Sent for maintenance
重物升至半空，停车后不能再起动。 When a heavy object is lifted to the halfway point, it cannot be started again after stopping	电压过低或波动大 Low voltage or large fluctuations	等电压恢复正常后再起动 Start after the voltage returns to normal
起动后，不能停车，或者到极限位置时，仍不能停车。 After start-up, it cannot be stopped, or when the limit position is reached, it also cannot be stopped.	交流接触器触头熔焊。 The contacts of the AC contactor are welded. 限位器失灵 Limiter malfunction	迅速切断总电源，拆卸检修或更换交流接触器 Quickly cut off the main power supply, remove for overhauling or replace the AC contactor
制动不可靠，下滑距离超过规定要求 The braking is unreliable, and the gliding distance exceeds the specified requirements	制动环与后端盖锥面接触不良 Poor contact between the brake ring and the cone of the rear end cover	拆下修磨 Remove for grinding
	制动面有油污 Oil fouling on the brake surface	拆下清洗 Remove for cleaning
	制动环松动 The brake ring is loose	更换制动环 Replace the brake ring
	压力弹簧疲劳 Pressure spring fatigue	更换弹簧 Replace spring
	联轴器窜动不灵或卡死 The coupling is not moving well or stuck	检查其联接部分 Check its connection part

表 6-3-4 桥式起重机故障及排除（大小车运行机构）

Table 6-3-4 Faults and Troubleshooting of Bridge Crane (Crane and Trolley Running Mechanisms)

故障名称 Fault Name		故障原因 Fault Cause	排除方法 Troubleshooting Method
起重机大车运行机构 Crane Running Mechanism	桥架歪斜运行、啃轨 The bridge runs skewed, gnawing the rails	1、两主动车轮直径误差过大 1. Excessive error in the diameter of the two driving wheels 2、主动车轮不是全部和轨道接触 2. Not all driving wheels are in contact with the track 3、主动轮轴线不正 3. Driving wheel axis misalignment 4、金属结构变形	1、测量、加工、更换车轮 1. Measuring, machining and replacing wheels 2、把满负荷小车开到大车落后的一端，如果大车走正，说明这端主动轮没和轨道全部接触，轮压小，可加大此端主动车轮直径 2. Drive the full load trolley to the backward end of the crane, if the crane is moving straight, it means that the driving wheel at this end is not fully in contact with the track, and the wheel pressure is small, so the diameter of the driving wheel at this end can be increased

		4. Deformation of metal structures 5、轨道安装质量差 5. Poor quality of track installation 6、轨顶有油污或冰霜 6. Oil fouling or frost on top of rails	3、检查和消除轴线偏斜现象 3. Check and eliminate axial skew phenomenon 4、矫正 4. Correct 5、调整轨道，使轨道符合安装技术条件 5. Adjust the track to meet the installation technical requirements 6、消除油污和冰霜 6. Eliminate oil fouling and frost
小车运行机构 Trolley running mechanism	打滑 Slip	1.轨顶有油污等 1. Oil fouling on top of rails 2、轮压不均 2. Uneven wheel pressure 3、同一截面内两轨道标高差过大 3. The elevation difference between the two tracks within the same section is too large 4、启、制动过于猛烈 4. The start and brake are too violent	1.清除 1. Remove 2.调整轮压 2. Adjust wheel pressure 3、调整轨道至符合技术条件 3. Adjust the track to meet the technical requirements 4、改善电动机启动方法，选用绕线式电动机 4. Improve the motor starting method and choose a wound-rotor induction motor
	启动时车身扭摆 Body roll when starting	1、小车轮压不均或主动车轮有一只悬空 1. Uneven pressure on the trolley's wheels or one of the driving wheels is suspended 2、啃轨 2. Gnawing rails	1、调整小车三条腿现象 1. Adjust the three-legged phenomenon of the trolley 2、解决啃轨 2. Solve track gnawing

设备名称 Equipment Name	垃圾搬运起重机 车 Waste handling crane	天气 Weather	
当班人： Person on duty:	接班人： Handover personnel:	编号 No.	

设备机构 Equipment Mechanism	工作内容 Working Contents	检查结果 Inspection Results	工作内容 Working Contents	检查结果 Inspection Results
移动供电 Mobile power supply	起重机牵引滑车 Crane towing pulley		小车牵引滑车 Trolley towing pulley	
	起重机电缆滑车 Crane cable pulley		小车电缆滑车 Trolley cable pulley	
	起重机供电电缆间距等 Crane power supply cable spacing, etc.		小车供电电缆间距等 Trolley power supply cable spacing, etc.	
	牵引绳锁间距等 Tow rope locking space, etc.		牵引绳锁间距等 Tow rope locking space, etc.	
电缆卷筒 Cable	垂直挂缆及上下接线盒 Vertical cable hanging and upper and lower junction box		滑环箱有无异常音响 Whether the slip ring box has abnormal sound	

drum				
	导线轮 Guide roller			
	电缆压板 Cable gland			
起 升 机构 Lifting mechanism	减速箱温度 Gearbox temperature		抓斗动作、声音 Grab actions, sounds	
	减速箱表面油污 Oil fouling on the surface of the gearbox		起重机上卫生纸状况 Sanitary conditions on cranes	
	制动器推杆伸出长度 Brake pushrod extension length			
	制动器温度 Brake temperature			
	制动瓦（单边）磨损情况 Wear condition of brake shoes (one side)			
	钢丝绳连接、破损情况 Wire rope connection and damage condition			
	钢丝绳卷筒轴 Wire rope drum shaft			
大 小 车 机 构 Crane and Trolley Mechanisms	三合一支撑点缓冲垫 Three-in-one support point buffer pad			
	三合一制动器动作 Three-in-one brake action			
	三合一减速箱温度 Three-in-one gearbox temperature			
填表人： Filled by:	归档人： Archived by:		日期： 年月日 Date: DD/MM/YYYY	
负责人： Person in charge:			日期： 年月日 Date: DD/MM/YYYY	

附件一、交接班情况记录单
Attachment 1. Handover Situation Record Sheet